

Projekt: Analiza ličnosti i njenih mračnih crta

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Motivacija i opis problema

Ličnost je jedno od temeljnih područja psihologije i najčešće se opisuje petofaktorskim Big Five modelom koji uključuje ekstraverziju, ugodnost, savjesnost, neuroticizam i otvorenost. Uz osnovne crte ličnosti, često se ispituju i mračne crte, odnosno mračna trijada koju čine narcizam, psihopatija i makijavelizam. Cilj ovog projekta je ispitati povezanost osnovnih i mračnih crta ličnosti, te ostalih obilježja poput spola i dobi, sa sklonošću stresu, depresiji i anksioznosti.

Učitavanje i pregled podataka

Podatke iz CSV datoteke učitavamo u varijablu *dataset* kako bismo ih mogle bolje analizirati.

```
dataset <- read_csv('Personality_data.csv')
```

```
## New names:
## Rows: 578 Columns: 18
## -- Column specification
## ----- Delimiter: "," chr
## (5): age, sex, ethnicity simplified, student status, employment status dbl
## (13): ...1, depression, anxiety, stress, narcissism, machiavelism, psych...
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'
```

Dalje, prikazemo prvih nekoliko redaka da vidimo primjere vrijednosti svakog od stupaca.

```
head(dataset)
```

```
## # A tibble: 6 x 18
##   ...1 depression anxiety stress narcissism machiavelism psychoticism sadism
##   <dbl>      <dbl>   <dbl> <dbl>      <dbl>      <dbl>      <dbl>   <dbl>
## 1     1      1.24    1.08  0.505      1.50      0.254     -0.201  0.0193
## 2     2      1.07    1.16  0.266      0.791      0.562      1.31   0.870
## 3     3     -0.581 -0.263 -0.556     -0.267     -0.692     -0.691 -0.208
## 4     4      3.03    1.99  1.94      0.415      0.856      2.19   1.10
## 5     5      1.22   -0.774  0.104     -2.34     -1.99     -0.733 -1.57
## 6     6      0.828  -0.186  0.728     -0.333      1.59      1.87   1.72
## # i 10 more variables: neuroticism <dbl>, extraversion <dbl>, openness <dbl>,
```

```
## # agreeableness <dbl>, conscientiousness <dbl>, age <chr>, sex <chr>,
## # 'ethnicity simplified' <chr>, 'student status' <chr>,
## # 'employment status' <chr>
```

Zatim, pomoću metoda `names()` i `glimpse()`, dajemo si bolji uvid u tipove podataka i strukturu.

```
names(dataset)
```

```
## [1] "...1"           "depression"       "anxiety"
## [4] "stress"          "narcissism"       "machiavelism"
## [7] "psychoticism"    "sadism"           "neuroticism"
## [10] "extraversion"    "openness"         "agreeableness"
## [13] "conscientiousness" "age"              "sex"
## [16] "ethnicity simplified" "student status"   "employment status"
```

```
glimpse(dataset)
```

```
## Rows: 578
## Columns: 18
## $ ...1          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ~
## $ depression    <dbl> 1.24028239, 1.07356352, -0.58072240, 3.03206310~
## $ anxiety       <dbl> 1.07785482, 1.15519418, -0.26310737, 1.98720356~
## $ stress        <dbl> 0.50512161, 0.26619388, -0.55579868, 1.94173788~
## $ narcissism     <dbl> 1.50455897, 0.79126351, -0.26693805, 0.41549063~
## $ machiavelism   <dbl> 0.25408691, 0.56182597, -0.69164422, 0.85587890~
## $ psychoticism   <dbl> -0.20141634, 1.31253355, -0.69076562, 2.1866752~
## $ sadism         <dbl> 0.01933759, 0.86951395, -0.20777818, 1.10452165~
## $ neuroticism    <dbl> -0.20124424, 0.56874269, -0.19237042, 1.4630829~
## $ extraversion   <dbl> 0.64878308, 0.44347482, -1.75862486, -1.8731478~
## $ openness       <dbl> 0.2455288, -0.9359119, -0.6614584, 1.8180786, --
## $ agreeableness  <dbl> -0.02353314, -1.17591066, -0.31643069, 0.655438~
## $ conscientiousness <dbl> -0.042944525, -1.043162336, -0.481776915, 0.039~
## $ age           <chr> "CONSENT_REVOKED", "CONSENT_REVOKED", "29", "37~
## $ sex           <chr> "CONSENT_REVOKED", "CONSENT_REVOKED", "Female",~
## $ 'ethnicity simplified' <chr> "CONSENT_REVOKED", "CONSENT_REVOKED", "Black", ~
## $ 'student status' <chr> "CONSENT_REVOKED", "CONSENT_REVOKED", "DATA_EXP~
## $ 'employment status' <chr> "CONSENT_REVOKED", "CONSENT_REVOKED", "DATA_EXP~
```

Prema uvidu u strukturu podataka vidljivo je da je varijabla **age** tipa *character*, što nije prikladno jer bi u statističkim modelima bila tretirana kao kategorijska umjesto numerička varijabla. Stoga je varijabla **age** pretvorena u numerički tip podataka.

Varijabla **sex** predstavlja kategorijsku varijablu te je, radi ispravne primjene u regresijskim modelima, pretvorena u tip *factor*.

Kako bi se osigurala valjanost analize, iz uzorka su uklonjeni svi ispitanici koji sadrže vrijednosti *CONSENT_REVOKED* ili *DATA_EXPIRED*, budući da ti zapisi ne predstavljaju stvarne podatke ispitanika.

```
data_clean <- dataset %>%
  filter(
    if_all(everything(),
      ~ !(. %in% c("CONSENT_REVOKED", "DATA_EXPIRED")))
  ) %>%
```

```

mutate(
  age = as.numeric(age),
  sex = factor(sex)
)

head(data_clean)

## # A tibble: 6 x 18
##   ...1 depression anxiety stress narcissism machiavelism psychoticism sadism
##   <dbl>         <dbl>   <dbl> <dbl>         <dbl>         <dbl>         <dbl> <dbl>
## 1     4         3.03     1.99  1.94         0.415         0.856         2.19  1.10
## 2     5         1.22    -0.774 0.104        -2.34        -1.99        -0.733 -1.57
## 3     6         0.828    -0.186 0.728        -0.333         1.59         1.87  1.72
## 4     7         0.0276   0.182 0.498         0.807         1.65         0.568 0.0681
## 5     9         0.633     1.14 0.421        -0.309         0.634         0.219 0.0423
## 6    10        -0.267    -1.68 0.188         0.608         0.294         0.442 0.780
## # i 10 more variables: neuroticism <dbl>, extraversion <dbl>, openness <dbl>,
## #   agreeableness <dbl>, conscientiousness <dbl>, age <dbl>, sex <fct>,
## #   'ethnicity simplified' <chr>, 'student status' <chr>,
## #   'employment status' <chr>

####Korelacijska matrica

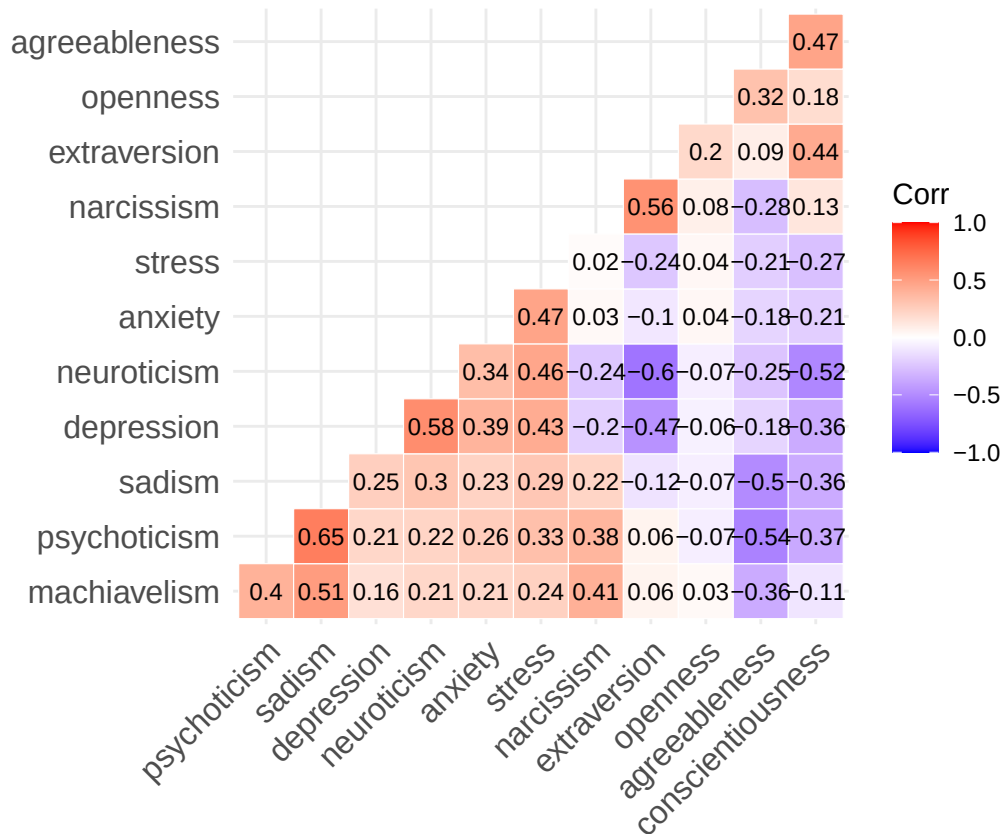
# odabir
traits_vars <- dataset %>%
  select(
    depression, anxiety, stress,
    narcissism, machiavelism, psychoticism, sadism,
    neuroticism, extraversion, openness, agreeableness, conscientiousness
  )

# korelacijska matrica
corr_mat <- cor(traits_vars, use = "pairwise.complete.obs")

# korelacijska heatmapa
ggcorrplot(
  corr_mat,
  type = "lower",
  lab = TRUE,
  lab_size = 3,
  hc.order = TRUE,
  outline.col = "white"
)

## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## i The deprecated feature was likely used in the ggcorrplot package.
## Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



Osobe s većom anksioznošću, depresijom i stresom imaju tendenciju biti i više neurotične.

Mračne osobine su međusobno umjereno povezane.

Visoka savjesnost štiti od depresije i stresa (negativna korelacija).

Ugodnost je snažno suprotna psihotizmu i sadizmu.

Big Five su međusobno slabo do umjereno povezane.

Najveće korelacije su:

Psychoticism–Sadism (0.65) Depression–Neuroticism (0.58) Extraversion–Narcissism(0.56) Machiavelism–Psychoticism (0.51)

Jesu li ljudi s izraženijom mračnom trijadom skloniji depresiji?

Upoznavanje podataka

Po analizi podataka već odrađenoj u uvodu znamo da radimo s numeričkim podacima, te da je za ispitivanje povezanosti osobina mračne trijade i depresije prikladno raditi linearnu regresiju. Dakle, nezavisne varijable ovdje su: narcizam, makijavelizam i psihotizam, a zavisna varijabla je depresija.

```
describe(data_clean %>%
  select(depression, narcissism, machiavelism, psychoticism))
```

```
##          vars    n  mean    sd median trimmed  mad   min   max range  skew
```

```
## depression      1 398  0.04 1.10 -0.14 -0.04 0.96 -3.07 3.63  6.70  0.63
## narcissism      2 398 -0.08 1.07  0.02 -0.05 1.16 -2.60 2.65  5.25 -0.16
## machiavelism    3 398 -0.05 1.08  0.05  0.00 1.09 -2.91 2.39  5.30 -0.37
## psychoticism    4 398 -0.08 1.00 -0.20 -0.14 1.04 -2.00 2.99  4.99  0.48
##               kurtosis  se
## depression      0.48 0.06
## narcissism     -0.63 0.05
## machiavelism   -0.41 0.05
## psychoticism   -0.48 0.05
```

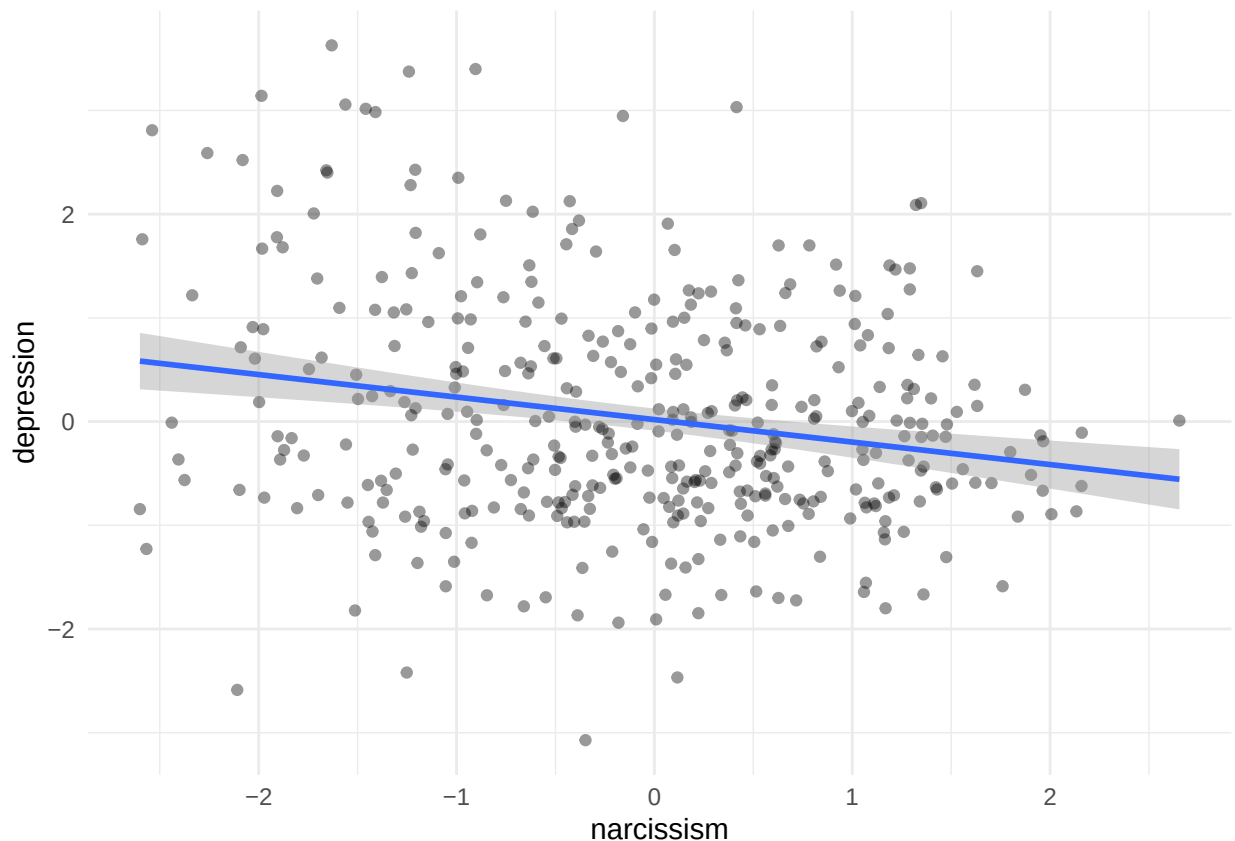
Vrijednosti svake od varijable kreću se negdje unutar intervala od -3.0 i 3.0, s medijanima i srednjim vrijednostima uvijek blizu nule. Također, uz $n=398$, znamo da radimo s velikim uzorkom što povećava statističku snagu analize.

Jednostavna linearna regresija

Kako bismo analizirali zaseban utjecaj svake od ovih varijabli, procijenit ćemo jednostavne regresijske modele, pri čemu će se za svaku nezavisnu varijablu izgraditi poseban model.

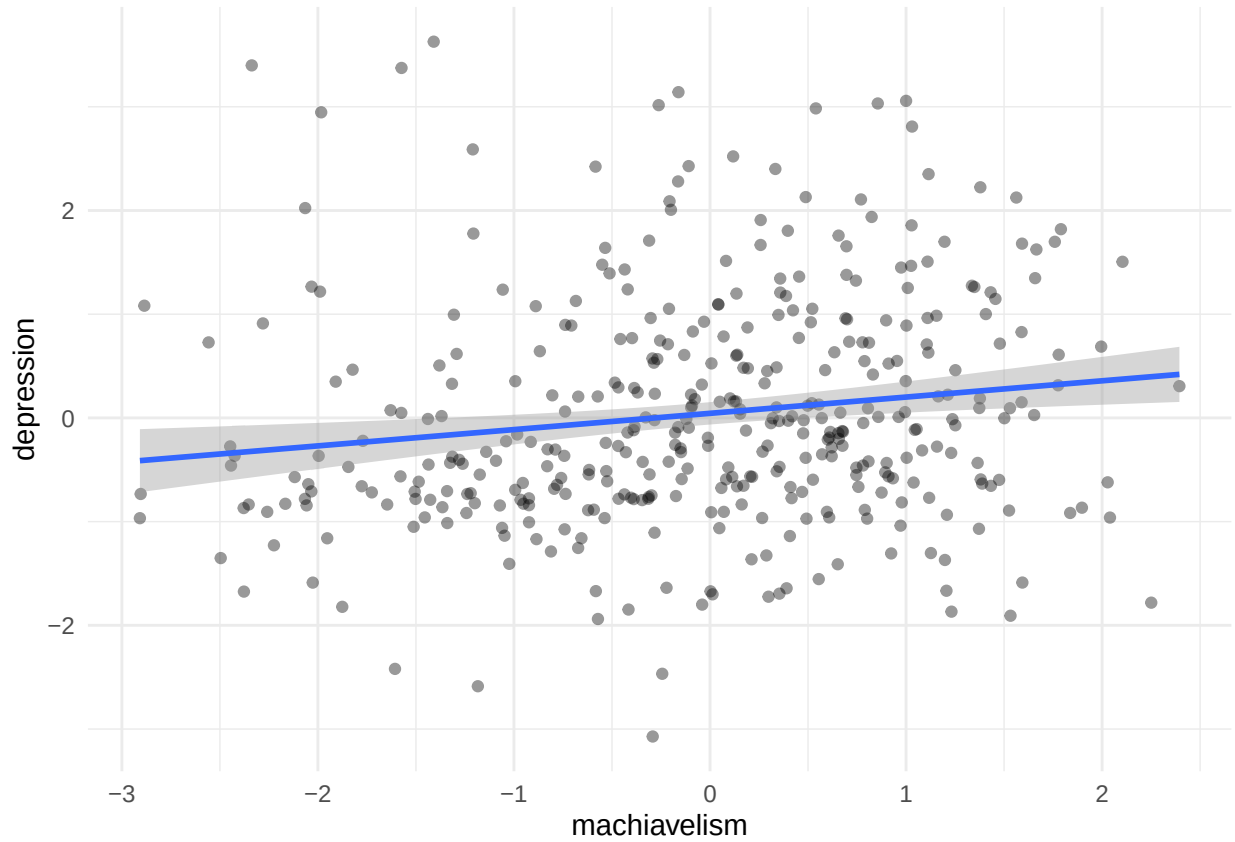
```
ggplot(data_clean, aes(narcissism, depression)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



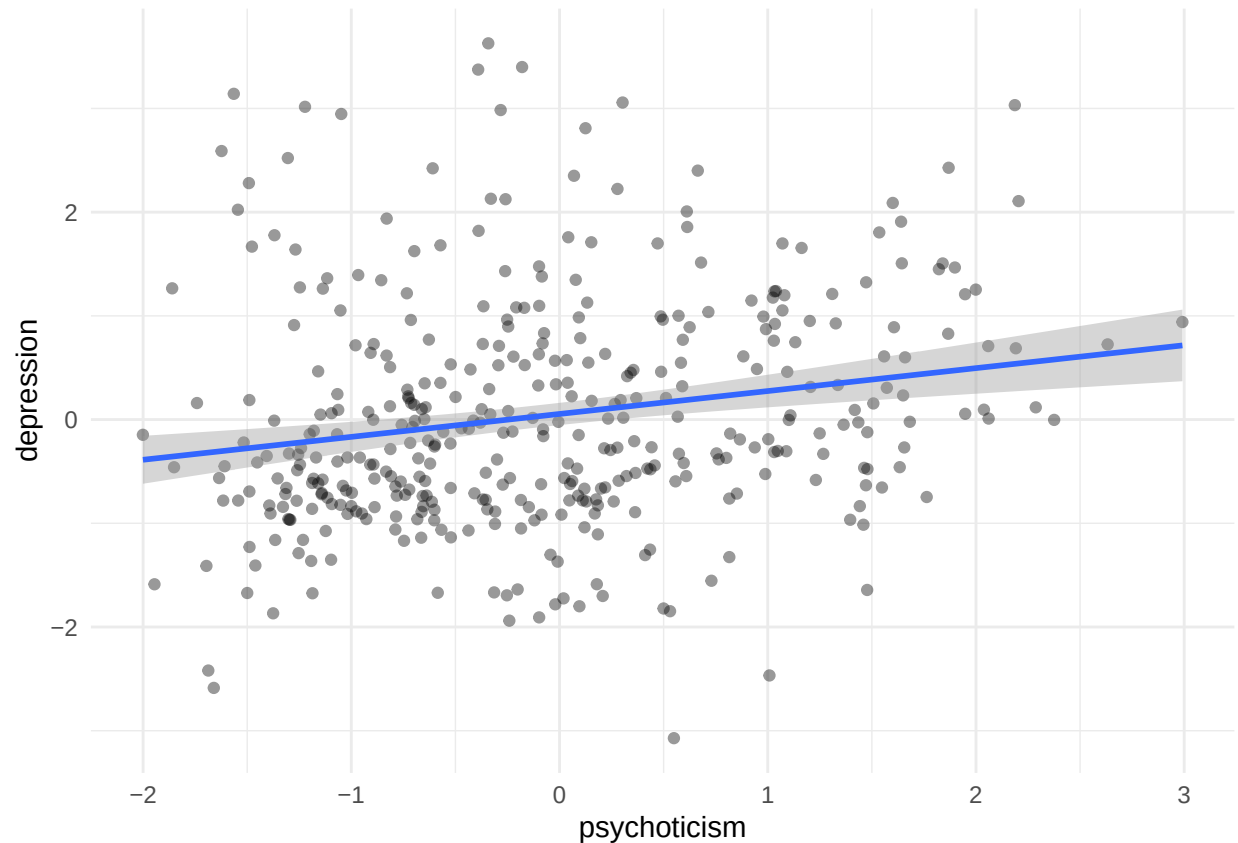
```
ggplot(data_clean, aes(machiavelism, depression)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggplot(data_clean, aes(psychoticism, depression)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Nagibi pravaca služe kao potvrda utjecaja pojedine nezavisne varijable na varijablu depresija. Model narcizma jedini ima negativan nagib, ukazujući na negativan utjecaj, dok psihoticizam i makijavelizam imaju pozitivan učinak.

Pearsonove korelacije

```
cor.test(data_clean$depression, data_clean$narcissism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$depression and data_clean$narcissism
## t = -4.2976, df = 396, p-value = 2.176e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.3031034 -0.1151869
## sample estimates:
## cor
## -0.2110948
```

```
cor.test(data_clean$depression, data_clean$machiavelism)
```

```
##
```

```
## Pearson's product-moment correlation
##
## data: data_clean$depression and data_clean$machiavelism
## t = 3.1025, df = 396, p-value = 0.002057
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.05660396 0.24857887
## sample estimates:
## cor
## 0.1540449
```

```
cor.test(data_clean$depression, data_clean$psychoticism)
```

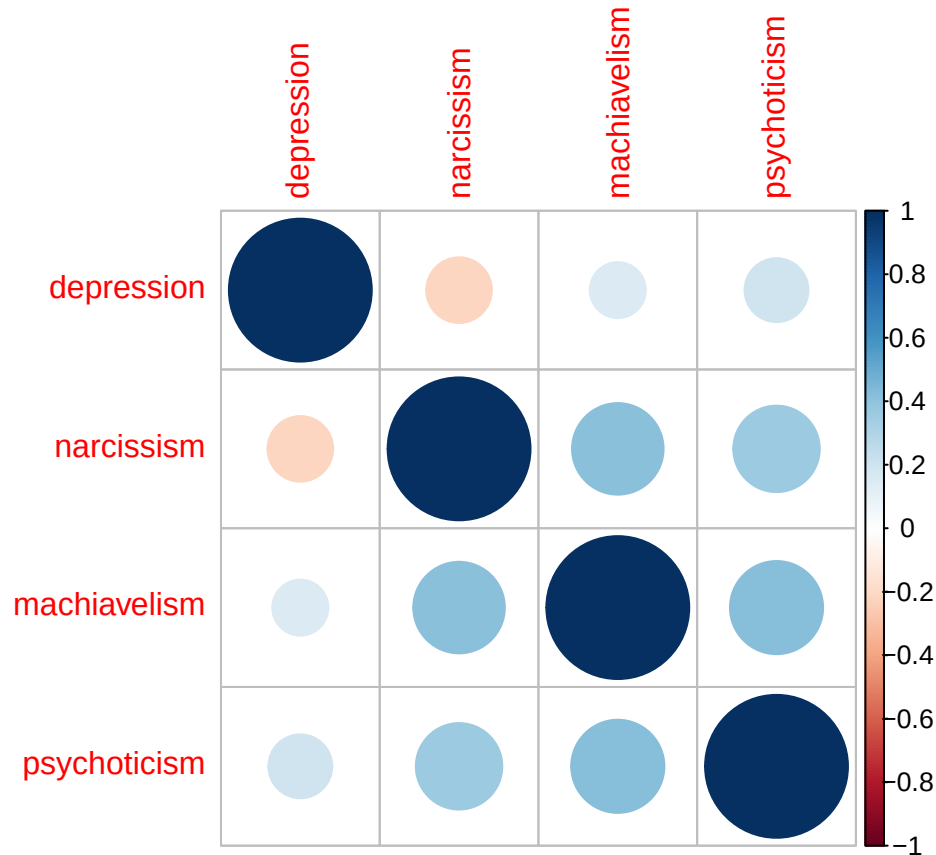
```
##
## Pearson's product-moment correlation
##
## data: data_clean$depression and data_clean$psychoticism
## t = 4.0662, df = 396, p-value = 5.767e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.1039454 0.2927352
## sample estimates:
## cor
## 0.2001979
```

Pearsonovom korelacijskom analizom utvrđena je statistički značajna negativna povezanost između narcizma i depresije ($r = -0.21$, $p < 0.001$). Makijavelizam ($r = 0.15$, $p = 0.002$) i psihoticizam ($r = 0.20$, $p < 0.001$) pokazali su statistički značajnu pozitivnu povezanost s depresijom. Veličine učinaka su male do umjerene.

Ovo se može prikazati i matrično:

```
vars1 <- data_clean %>%
  select(depression, narcissism, machiavelism, psychoticism) %>%
  na.omit()

corrplot(cor(vars1))
```

Među prediktorima su prisutne umjerene pozitivne korelacije, no nijedna ne upućuje na problematičnu multikolinearnost. Ovo ćemo još formalno analizirati koristeći, ponovno, Pearsonove korelacije.

```
cor.test(data_clean$machiavelism, data_clean$narcissism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$machiavelism and data_clean$narcissism
## t = 8.961, df = 396, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3254326 0.4891514
## sample estimates:
## cor
## 0.4105959
```

```
cor.test(data_clean$machiavelism, data_clean$psychoticism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$machiavelism and data_clean$psychoticism
## t = 9.3422, df = 396, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
```

```
## 95 percent confidence interval:
## 0.3409053 0.5022790
## sample estimates:
##      cor
## 0.4249627
```

```
cor.test(data_clean$narcissism, data_clean$machiavelism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$narcissism and data_clean$machiavelism
## t = 8.961, df = 396, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3254326 0.4891514
## sample estimates:
##      cor
## 0.4105959
```

```
cor.test(data_clean$narcissism, data_clean$psychoticism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$narcissism and data_clean$psychoticism
## t = 7.8298, df = 396, p-value = 4.519e-14
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2778423 0.4483037
## sample estimates:
##      cor
## 0.3661405
```

```
cor.test(data_clean$psychoticism, data_clean$narcissism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$psychoticism and data_clean$narcissism
## t = 7.8298, df = 396, p-value = 4.519e-14
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2778423 0.4483037
## sample estimates:
##      cor
## 0.3661405
```

```
cor.test(data_clean$psychoticism, data_clean$machiavelism)
```

```
##
```

```
## Pearson's product-moment correlation
##
## data: data_clean$psychoticism and data_clean$machiavelism
## t = 9.3422, df = 396, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3409053 0.5022790
## sample estimates:
## cor
## 0.4249627
```

Pearsonove korelacije između nezavisnih varijabli pokazale su umjerene pozitivne povezanosti između makijavelizma, narcizma i psihotizma ($r \approx 0,36-0,42$, $p < 0,001$). Iako su ove osobine povezane, snaga korelacija ne sugerira problematičnu multikolinearnost, što podupire njihovo uključivanje kao zasebnih prediktora u regresijski model.

Višestruka linearna regresija

Sada možemo definirati model višestruke linearne regresije, gdje gledamo sveukupni utjecaj mračne trijade na depresiju.

```
model_dark_triad <- lm(depression ~ narcissism + machiavelism + psychoticism,
                        data = data_clean)

summary(model_dark_triad)
```

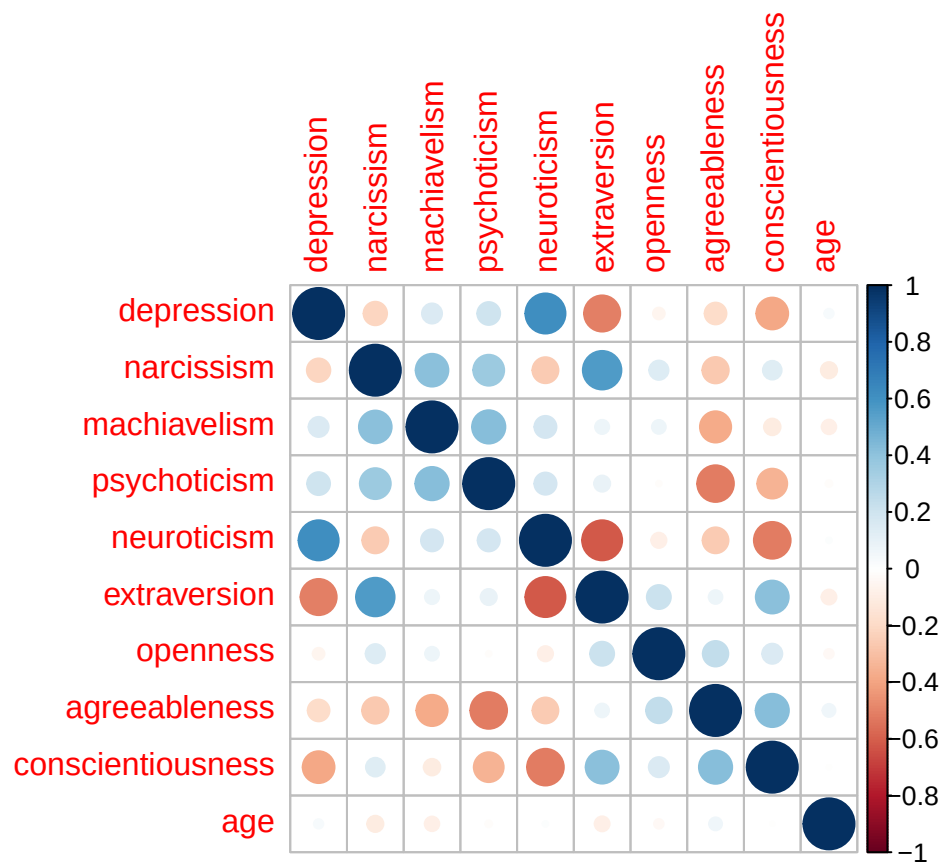
```
##
## Call:
## lm(formula = depression ~ narcissism + machiavelism + psychoticism,
##     data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3437 -0.6086 -0.1376  0.5302  3.5540
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.03894    0.05100   0.763 0.445638
## narcissism   -0.39957    0.05334  -7.490 4.55e-13 ***
## machiavelism  0.20893    0.05434   3.845 0.000141 ***
## psychoticism  0.28178    0.05770   4.884 1.52e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.013 on 394 degrees of freedom
## Multiple R-squared:  0.1648, Adjusted R-squared:  0.1585
## F-statistic: 25.92 on 3 and 394 DF, p-value: 2.545e-15
```

Rezultati reiteriraju prijašnje zaključke da je narcisizam negativno značajan prediktor depresije ($\beta = -0.40$, $p < 0.001$), dok su makijavelizam ($\beta = 0.21$, $p < 0.001$) i psihotizam ($\beta = 0.28$, $p < 0.001$) pozitivno značajni prediktori. Model je statistički značajan (F-stat = 25.92, $p < 0.001$). Međutim, ukupna prediktivna snaga modela je umjerena (R^2 oko 0.16), što sugerira da postoje i drugi faktori koji utječu na depresiju.

Kako bismo provjerili postoje li drugi, prikladniji i značajniji, prediktori prikazat ćemo korelacijsku matricu sa ostalim varijablama.

```
vars2 <- data_clean %>%
  select(depression, narcissism, machiavelism, psychoticism,
         neuroticism, extraversion, openness,
         agreeableness, conscientiousness,
         age) %>%
  na.omit()

corrplot(cor(vars2))
```



Tu su vidljive pozitivne povezanosti depresije s neuroticizmom, psihoticizmom i najmanje makijavelizmom. S druge strane, negativnu korelaciju s depresijom imaju narcizam, ekstraverzija i savjesnost, nešto manje kod ugodnosti. Najjače povezanosti su sa ekstraverzijom i neuroticizmom, koji su međusobno negativno korelirani.

```
model_2 <- lm(depression ~ narcissism + extraversion + neuroticism + conscientiousness + agreeableness +
              data = data_clean)

summary(model_2)
```

```
##
## Call:
## lm(formula = depression ~ narcissism + extraversion + neuroticism +
```

```
##      conscientiousness + agreeableness + psychoticism, data = data_clean)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -3.0698 -0.4943  0.0391  0.4561  2.9942
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.017541   0.042738   0.410  0.68172
## narcissism     -0.007888   0.053502  -0.147  0.88287
## extraversion   -0.238504   0.060134  -3.966 8.68e-05 ***
## neuroticism     0.456224   0.055935   8.156 4.76e-15 ***
## conscientiousness -0.030049  0.055976  -0.537  0.59170
## agreeableness    0.038033   0.052069   0.730  0.46556
## psychoticism    0.169456   0.054041   3.136  0.00184 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8458 on 391 degrees of freedom
## Multiple R-squared:  0.4222, Adjusted R-squared:  0.4133
## F-statistic: 47.61 on 6 and 391 DF,  p-value: < 2.2e-16
```

Model značajno predviđa depresiju ($R^2 = 0.42$, $p < 0.001$). Neuroticizam je bio najsnažniji pozitivni prediktor depresije ($\beta = 0.46$, $p < 0.001$), dok je ekstraverzija imala značajan negativan učinak ($\beta = -0.24$, $p < 0.001$). Psihoticizam je također pokazao umjeren, ali značajan pozitivan doprinos ($\beta = 0.17$, $p = 0.0018$). Narcizam, savjesnost i ugodnost nisu bili statistički značajni prediktori ($p > 0.05$), što upućuje na to da njihov odnos s depresijom nije jedinstven nakon kontrole ostalih osobina.

Još ćemo provjeriti predstavlja li kolinearnost značajki problem za ovu interpretaciju:

```
vif <- function(model) {
  X <- model.matrix(model)[, -1]
  vif_values <- sapply(1:ncol(X), function(i) {
    lm_fit <- lm(X[, i] ~ X[, -i])
    1 / (1 - summary(lm_fit)$r.squared)
  })
  names(vif_values) <- colnames(X)
  return(vif_values)
}

vif(model_2)
```

```
##      narcissism      extraversion      neuroticism conscientiousness
##      1.834314        2.376552        1.930409        1.706299
##      agreeableness      psychoticism
##      1.598789          1.623147
```

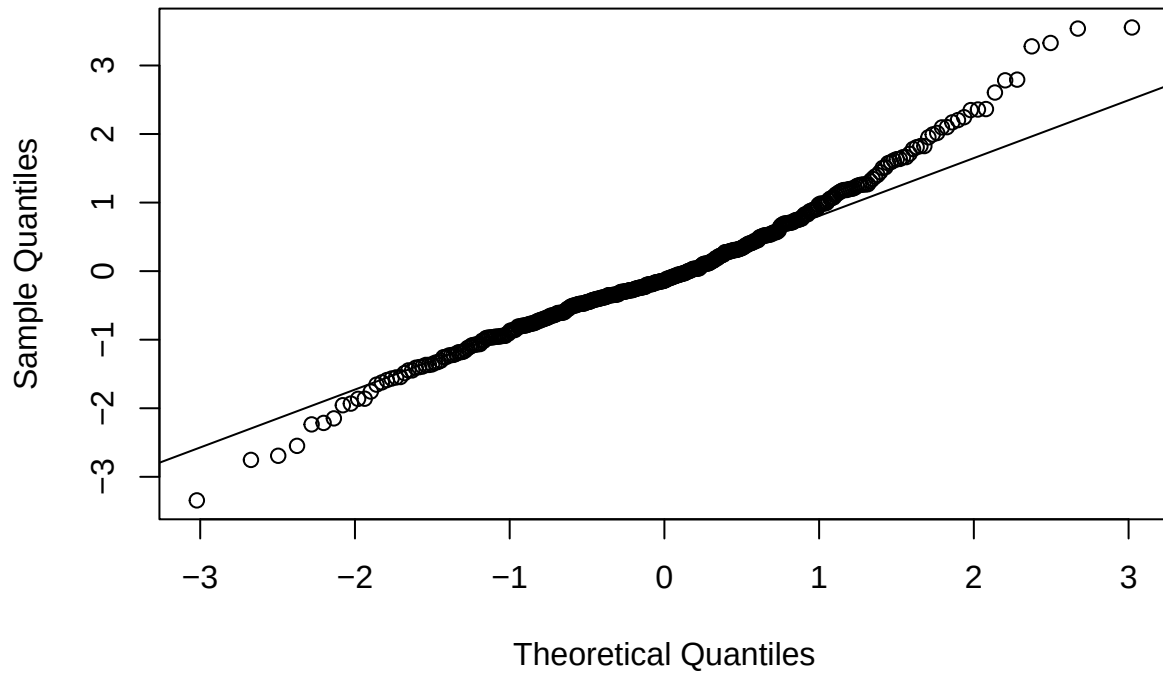
VIF faktori za sve prediktore manji su od 5, što ukazuje da ne postoji značajna multikolinearnost među varijablama.

Provjera pretpostavki linearnih regresija

Još je preostalo provjeriti prate li reziduali ovih modela normalnu distribuciju.

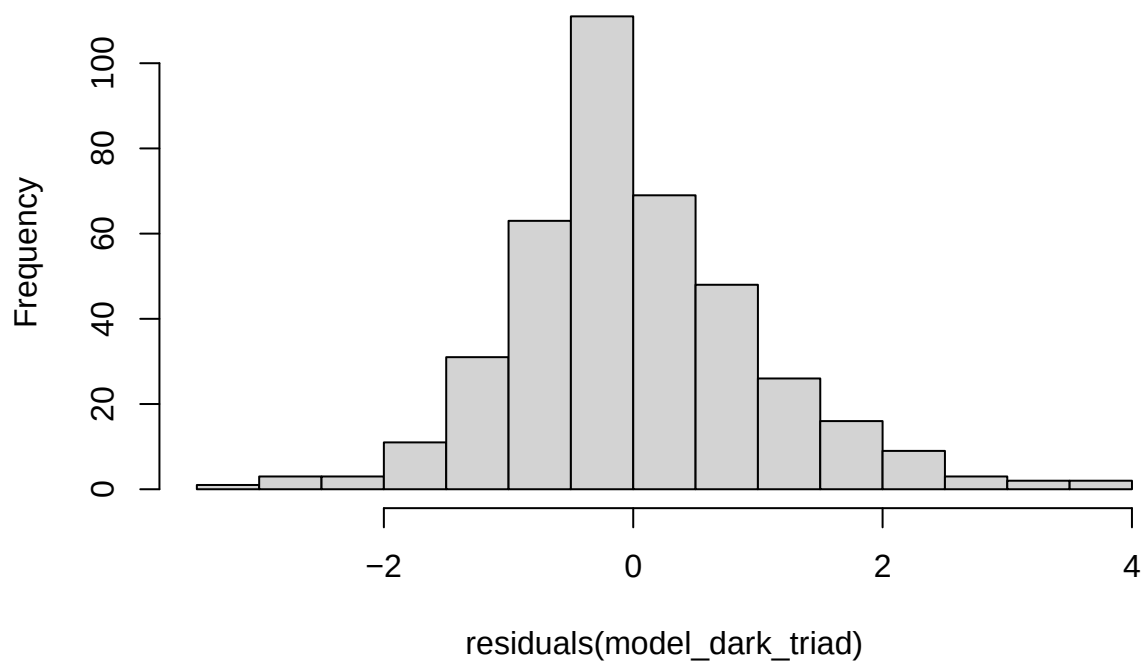
```
qqnorm(
  residuals(model_dark_triad),
  main = "QQ-plot reziduala modela depresije i mračne trijade"
)
qqline(residuals(model_dark_triad))
```

QQ-plot reziduala modela depresije i mračne trijade



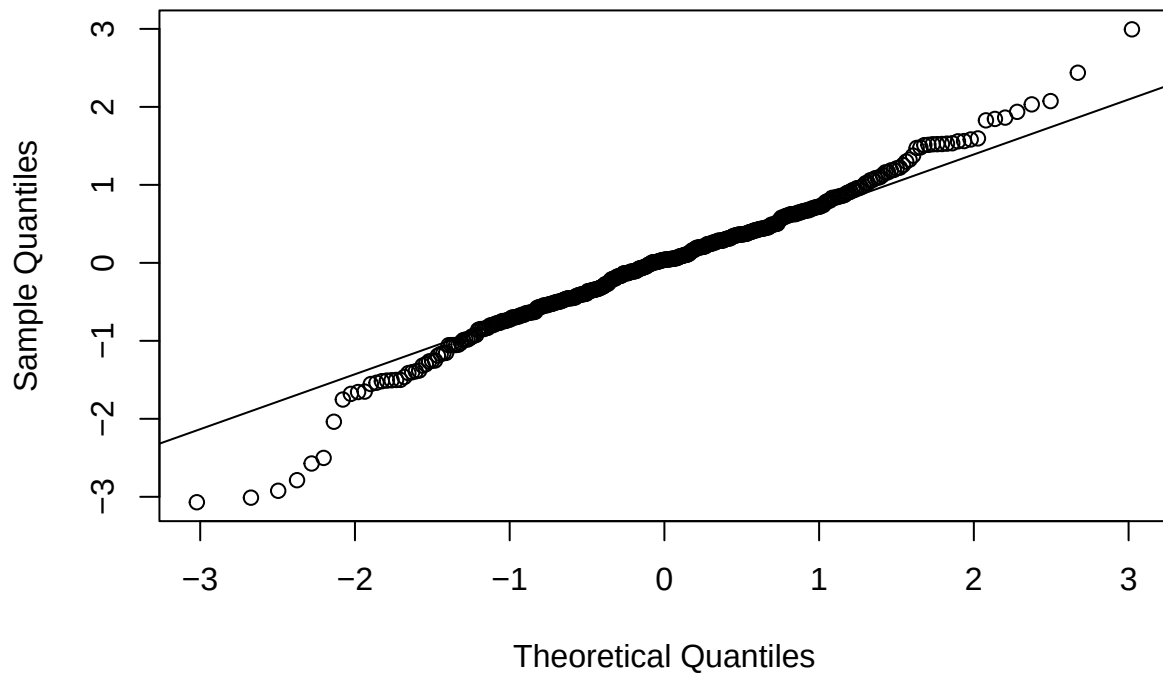
```
hist(residuals(model_dark_triad), breaks = 20, main = "Histogram reziduala modela depresije i mračne tr
```

Histogram reziduala modela depresije i mracne trijade



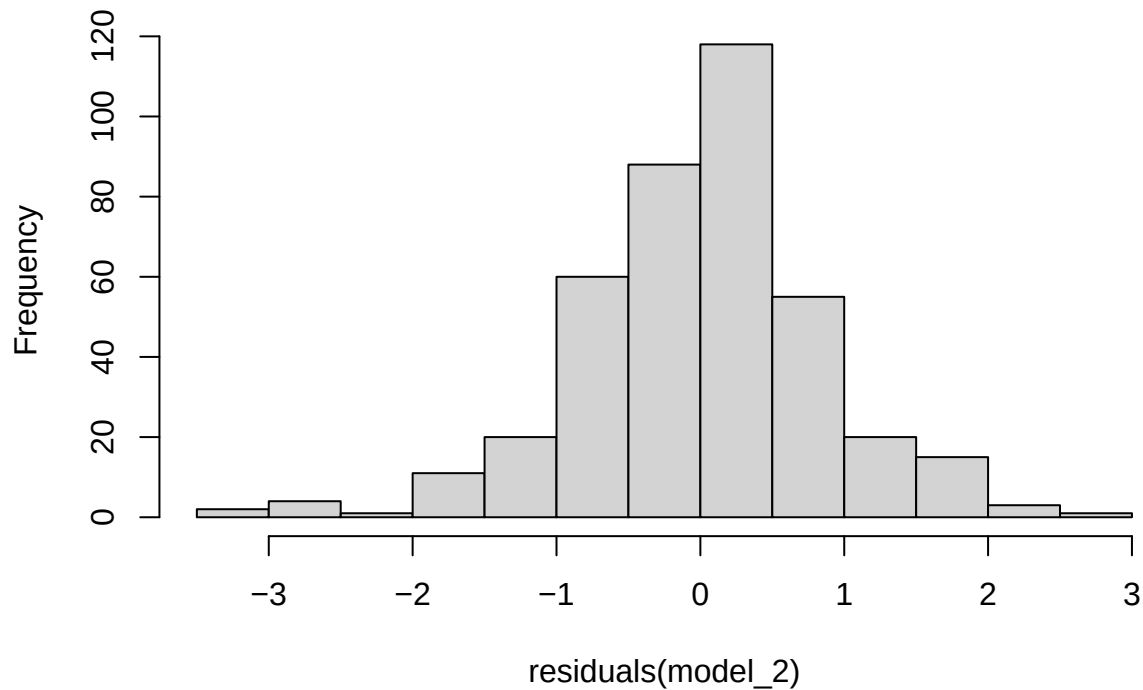
```
qqnorm(  
  residuals(model_2),  
  main = "QQ-plot reziduala modela depresije i ostalih značajki"  
)  
qqline(residuals(model_2))
```

QQ-plot reziduala modela depresije i ostalih znacajki



```
hist(residuals(model_2), breaks = 20, main = "Histogram reziduala modela depresije i ostalih znacajki")
```


Histogram reziduala modela depresije i ostalih znacajki



Normalnost reziduala provjerena je pomoću QQ-plota i histograma, koji su pokazali približno normalnu distribuciju, uz manja odstupanja na krajevima distribucije. Odstupanja su zastupljenija kod drugog modela koji ne uključuje samo varijable mračne trijade, već i ostale značajke.

Zaključak

Pokazali smo da osobine koje čine mračnu trijadu: psihoticizam, makijavelizam i narcizam, same po sebi utječu na depresiju, ali njihov učinak nije jednako snažan kao učinak nekih drugih osobina, poput niske ekstraverzije i visokog neuroticizma s elementima psihoticizma.

Postoje li razlike u nekim crtama ličnosti među spolovima?

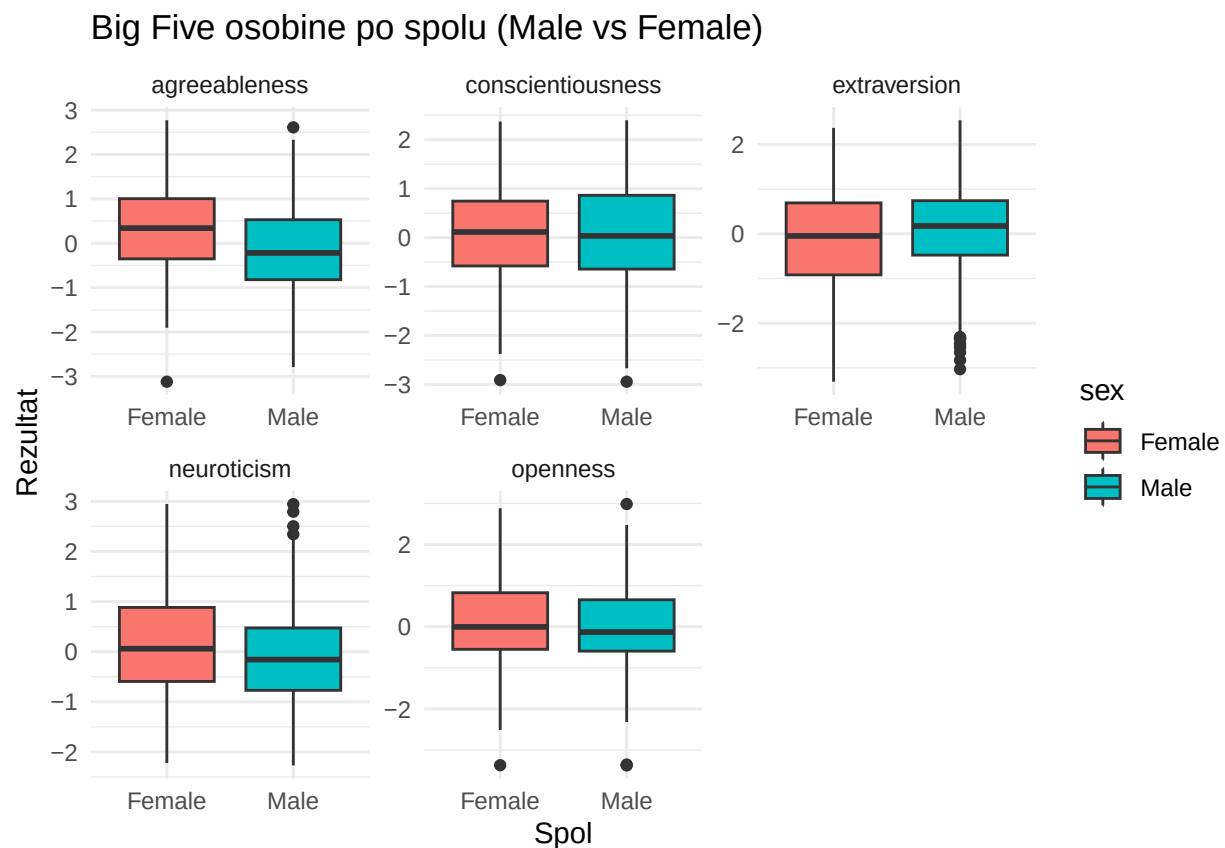
```
table(data_clean$sex)
```

```
##  
##           Female           Male Prefer not to say  
##           186           209             3
```

```
df_sex <- data_clean %>%  
  filter(sex %in% c("Male", "Female"))
```

```
big_five <- df_sex %>%
  pivot_longer(
    cols = c(neuroticism, extraversion, openness, agreeableness, conscientiousness),
    names_to = "trait",
    values_to = "score"
  )

ggplot(big_five, aes(x = sex, y = score, fill = sex)) +
  geom_boxplot() +
  facet_wrap(~ trait, scales = "free") +
  theme_minimal() +
  labs(
    title = "Big Five osobine po spolu (Male vs Female)",
    x = "Spol",
    y = "Rezultat"
  )
```



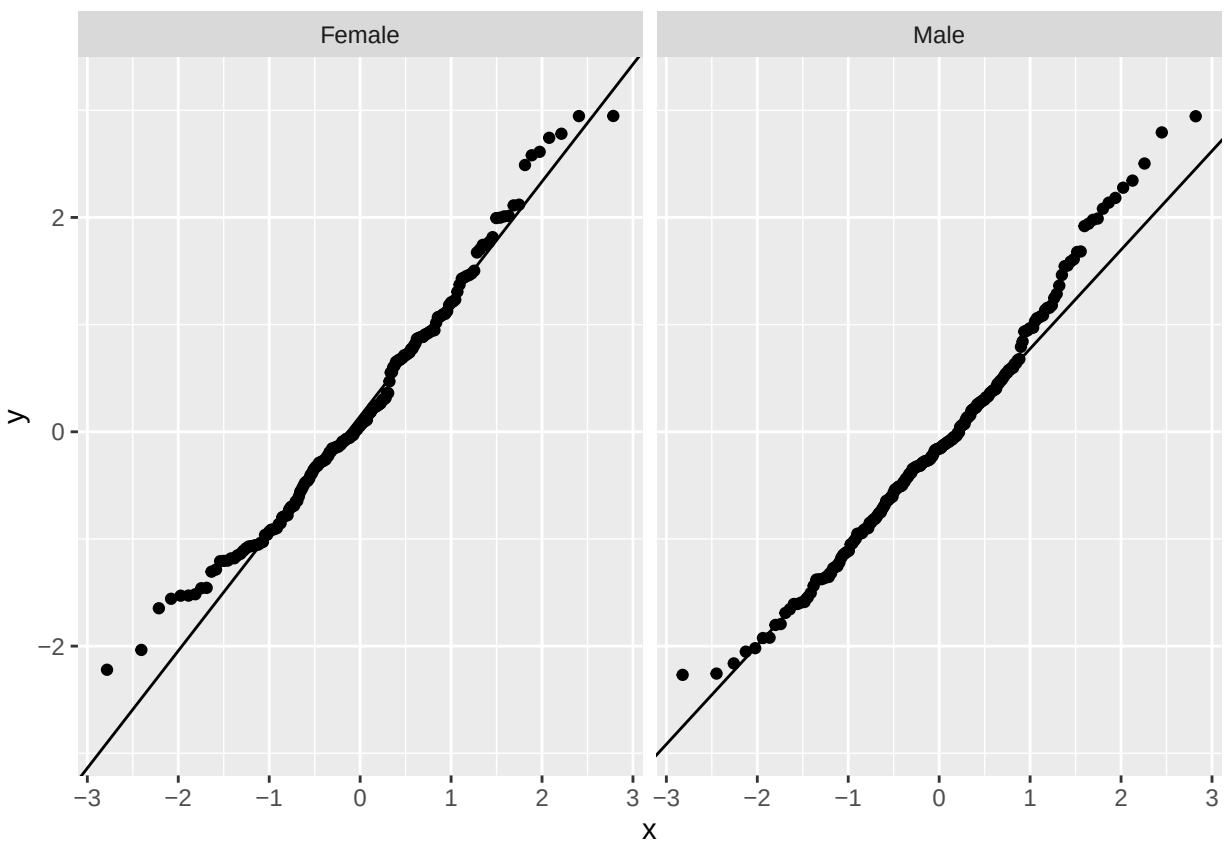
Vizualna analiza boxplotova ukazuje na obrasce razlika između spolova koji su u skladu s teorijskim očekivanjima. Najizraženije razlike uočavaju se kod neuroticizma i ugodnosti, pri čemu žene u prosjeku postižu više vrijednosti, što je vidljivo kroz više medijane i pomak distribucije prema višim vrijednostima.

Kod ekstraverzije, otvorenosti i savjesnosti razlike između spolova su manje izražene; medijani su vrlo slični, a distribucije se u velikoj mjeri preklapaju, što upućuje na izostanak izraženih spolnih razlika u tim osobinama na razini deskriptivne analize.

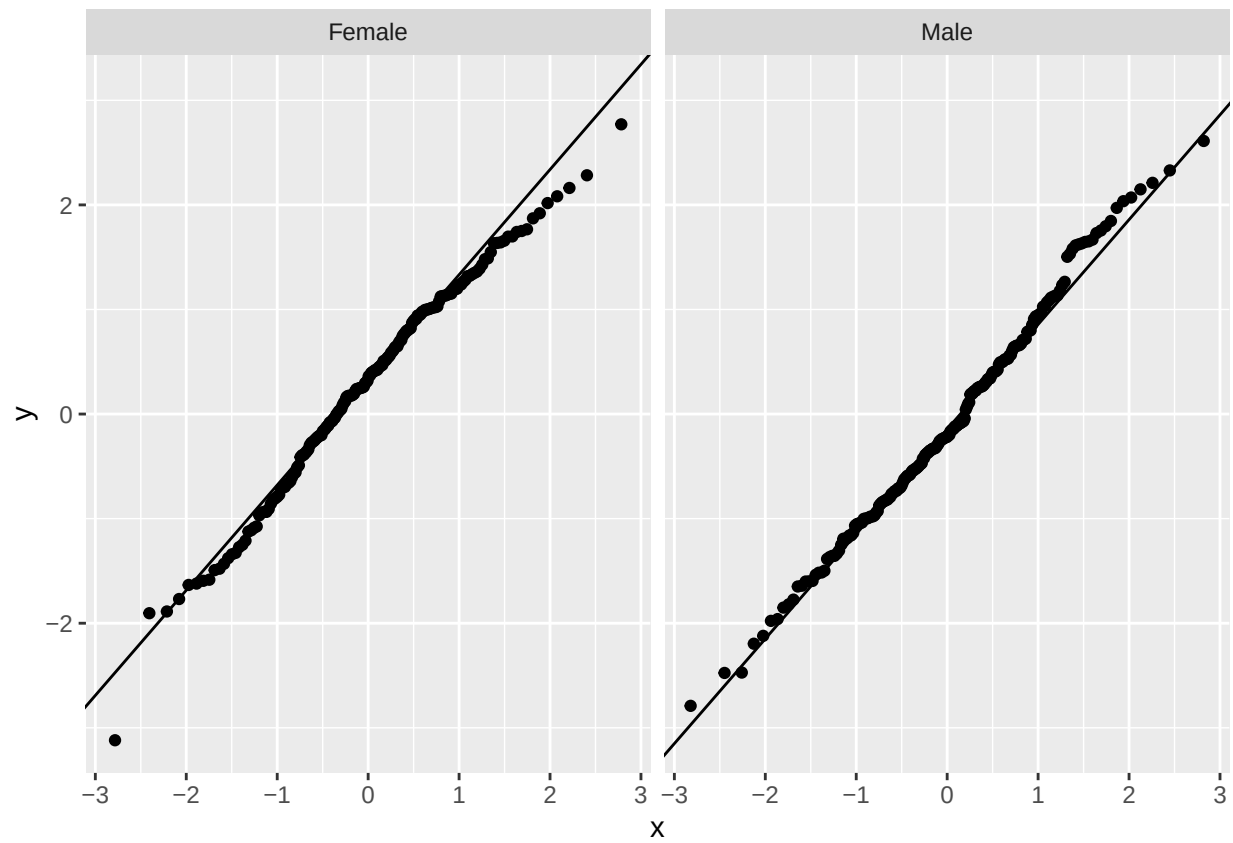
Prije provedbe t-testa za nezavisne uzorke, potrebno je provjeriti jesu li zadovoljene osnovne pretpostavke testa, uključujući normalnost distribucije, homogenost varijanci te nezavisnost uzoraka.

Za početak ćemo provjeriti pretpostavku o normalnosti uzorka.

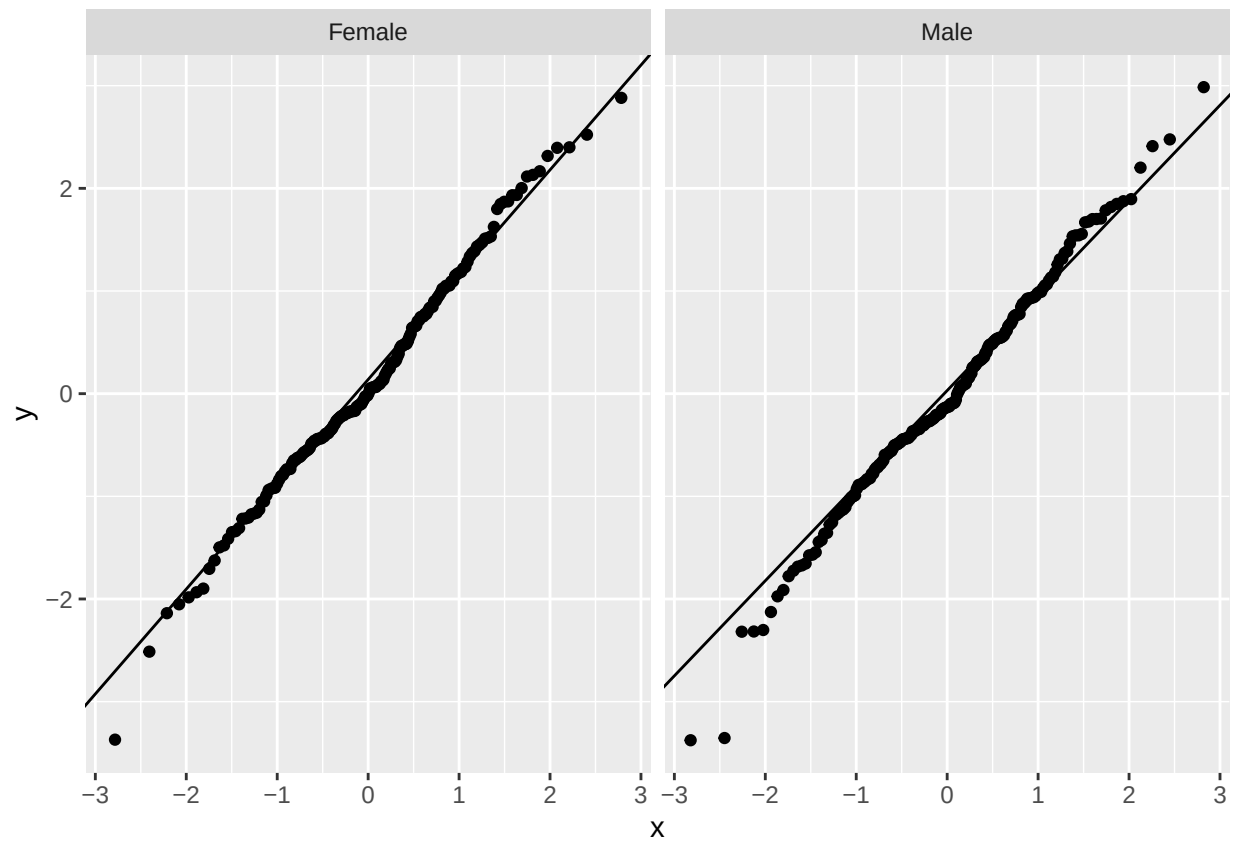
```
ggplot(df_sex, aes(sample = neuroticism)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



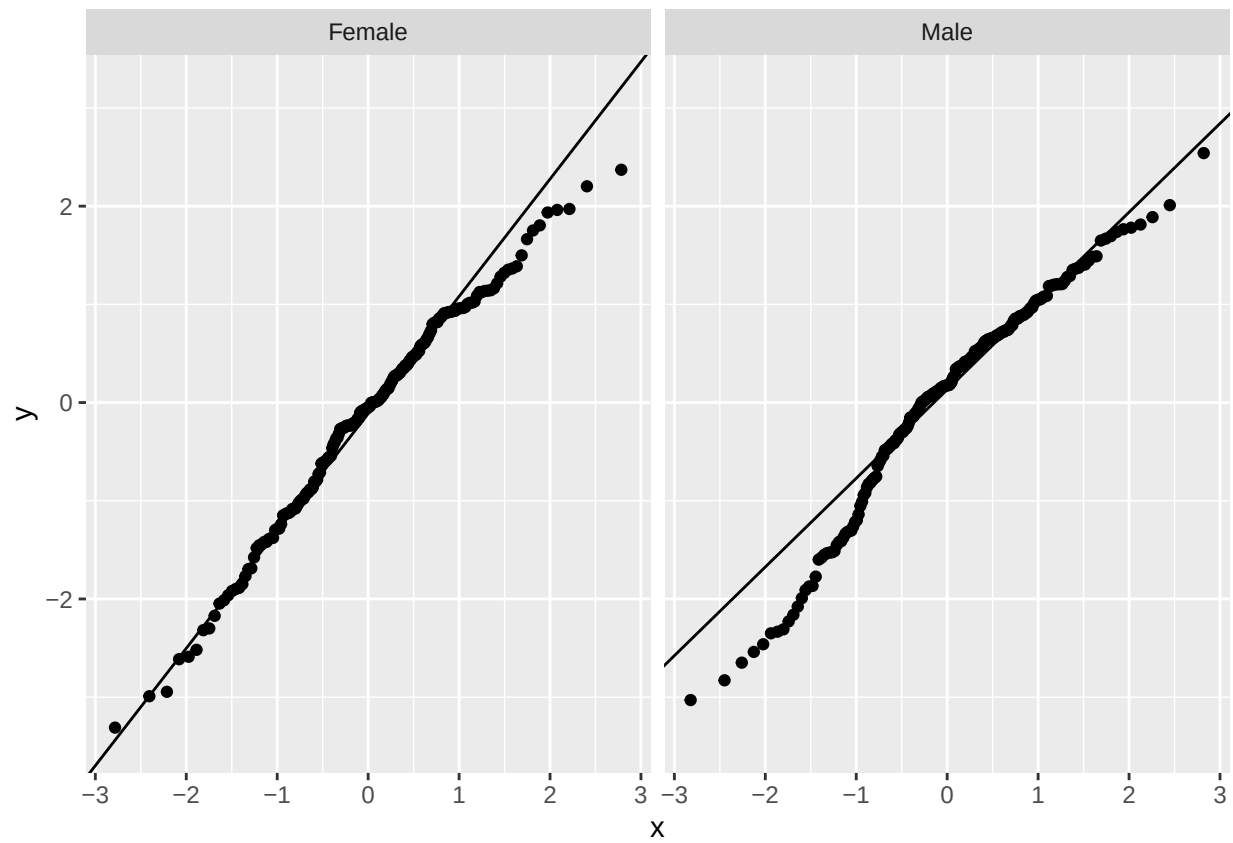
```
ggplot(df_sex, aes(sample = agreeableness)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



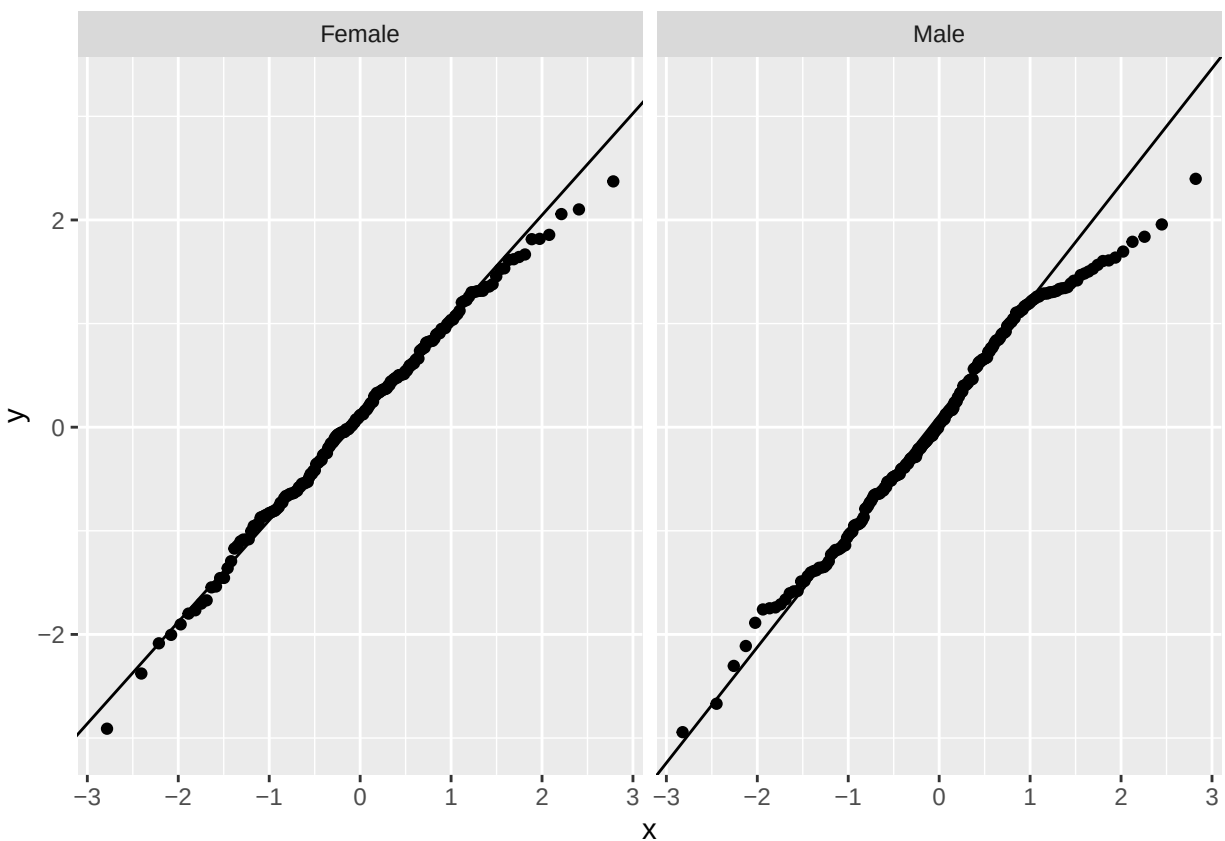
```
ggplot(df_sex, aes(sample = openness)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



```
ggplot(df_sex, aes(sample = extraversion)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



```
ggplot(df_sex, aes(sample = conscientiousness)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



Distribucije u skupinama su približno normalne. Zatim ćemo provjeriti varijance.

```
traits <- c("neuroticism", "agreeableness", "openness", "extraversion", "conscientiousness")

var_by_sex <- df_sex %>%
  group_by(sex) %>%
  summarise(across(all_of(traits), ~ var(.x, na.rm = TRUE)))

var_by_sex
```

```
## # A tibble: 2 x 6
##   sex      neuroticism agreeableness openness extraversion conscientiousness
##   <fct>          <dbl>          <dbl>    <dbl>         <dbl>          <dbl>
## 1 Female          1.11            0.977    1.13          1.23          0.897
## 2 Male            1.06            1.07    1.08          1.14          1.03
```

```
var_tests <- lapply(traits, function(tr) {
  f <- as.formula(paste(tr, "~ sex"))
  vt <- var.test(f, data = df_sex)
  tibble(
    trait = tr,
    F_statistic = round(vt$statistic, 3),
    p_value = round(vt$p.value, 4)
  )
})
```

```
bind_rows(var_tests)
```

```
## # A tibble: 5 x 3
##   trait          F_statistic p_value
##   <chr>          <dbl>    <dbl>
## 1 neuroticism      1.05    0.740
## 2 agreeableness    0.916    0.543
## 3 openness         1.05    0.752
## 4 extraversion     1.08    0.606
## 5 conscientiousness 0.875    0.354
```

Rezultati testa za jednakost varijanci pokazuju da ni za jednu od analiziranih osobina ličnosti nije utvrđena statistički značajna razlika u varijancama između muškaraca i žena. Sve dobivene p-vrijednosti veće su od 0.05, stoga se u svim slučajevima ne odbacuje nul-hipoteza o jednakosti varijanci.

Dodatno, promatramo dvije nezavisne skupine: Female i Male (ostale ćemo ukloniti).

Stoga, sve pretpostavke su zadovoljene i provest ćemo t-test za nezavisne uzorke.

```
# t-test za svaku osobinu
results <- lapply(trait, function(tr) {
  formula <- as.formula(paste(tr, "~ sex")) #npr. neuroticism ~ sex
  test <- t.test(formula, data = df_sex)
  print(test$estimate)
  tibble(
    trait = tr, #naziv osobine
    t_statistic = round(test$statistic, 3), #t vrijednost
    p_value = round(test$p.value, 5),
    mean_female = round(test$estimate["mean in group Female"], 3),
    mean_male = round(test$estimate["mean in group Male"], 3)
  )
})
```

```
## mean in group Female mean in group Male
##      0.18251516      -0.08261147
## mean in group Female mean in group Male
##      0.2738572      -0.1091680
## mean in group Female mean in group Male
##      0.10694830      -0.02212707
## mean in group Female mean in group Male
##      -0.14239955      0.04568951
## mean in group Female mean in group Male
##      0.07941870      0.03722745
```

```
results_df <- bind_rows(results)
results_df
```

```
## # A tibble: 5 x 5
##   trait          t_statistic p_value mean_female mean_male
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 neuroticism      2.52  0.0120      0.183    -0.083
## 2 agreeableness    3.76  0.00019      0.274    -0.109
```


## 3 openness	1.22	0.223	0.107	-0.022
## 4 extraversion	-1.71	0.0873	-0.142	0.046
## 5 conscientiousness	0.428	0.669	0.079	0.037

Rezultati dvostranih t-testova za nezavisne uzorke ukazuju na postojanje statistički značajnih spolnih razlika u pojedinim dimenzijama ličnosti. Žene postižu značajno više prosječne rezultate od muškaraca u neuroticizmu ($t = 2.52$, $p = 0.012$) i ugodnosti ($t = 3.76$, $p < 0.001$).

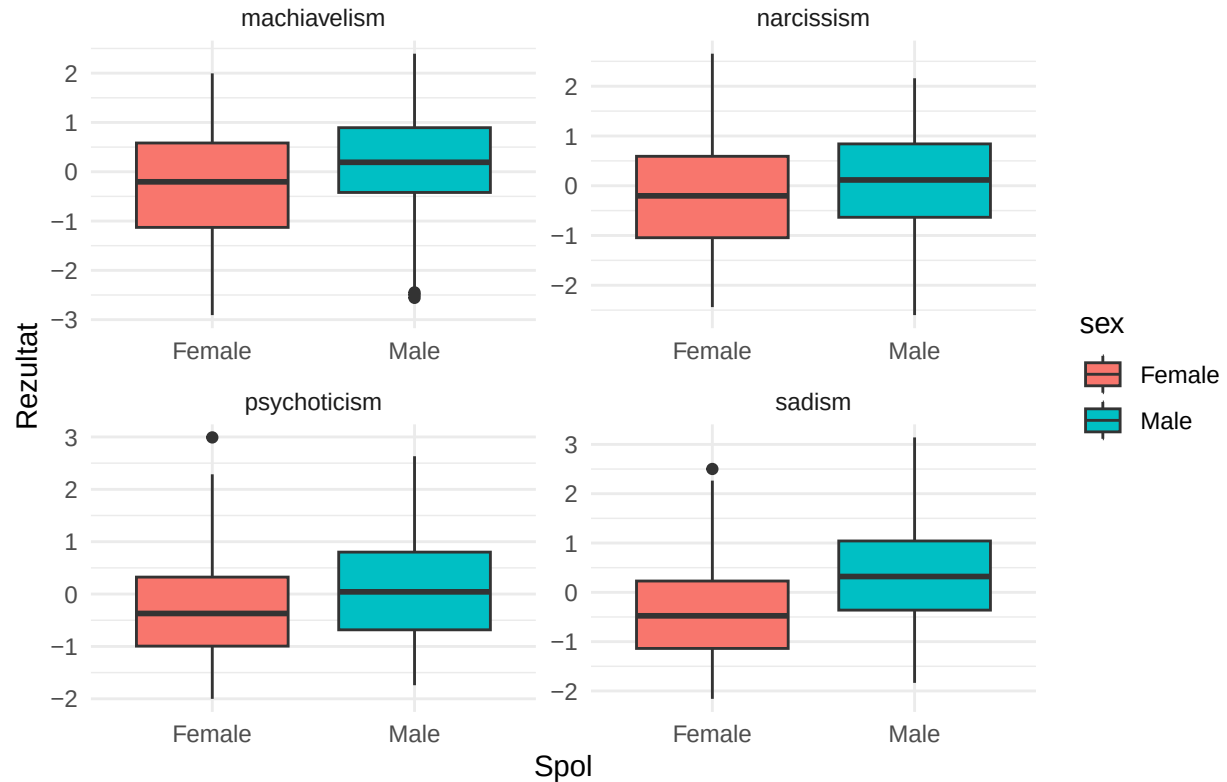
Za otvorenost ($t = 1.22$, $p = 0.223$), ekstraverziju ($t = -1.71$, $p = 0.087$) i savjesnost ($t = 0.43$, $p = 0.669$) nisu utvrđene statistički značajne razlike između spolova, iako su uočene blage razlike u prosječnim vrijednostima.

Sad ćemo pogledati mračne osobine.

```
dark_traits <- df_sex %>%
  pivot_longer(
    cols = c(narcissism, machiavelism, psychoticism, sadism),
    names_to = "trait",
    values_to = "score"
  )

ggplot(dark_traits, aes(x = sex, y = score, fill = sex)) +
  geom_boxplot() +
  facet_wrap(~ trait, scales = "free") +
  theme_minimal() +
  labs(
    title = "Mračne osobine ličnosti po spolu (Male vs Female)",
    x = "Spol",
    y = "Rezultat"
  )
```

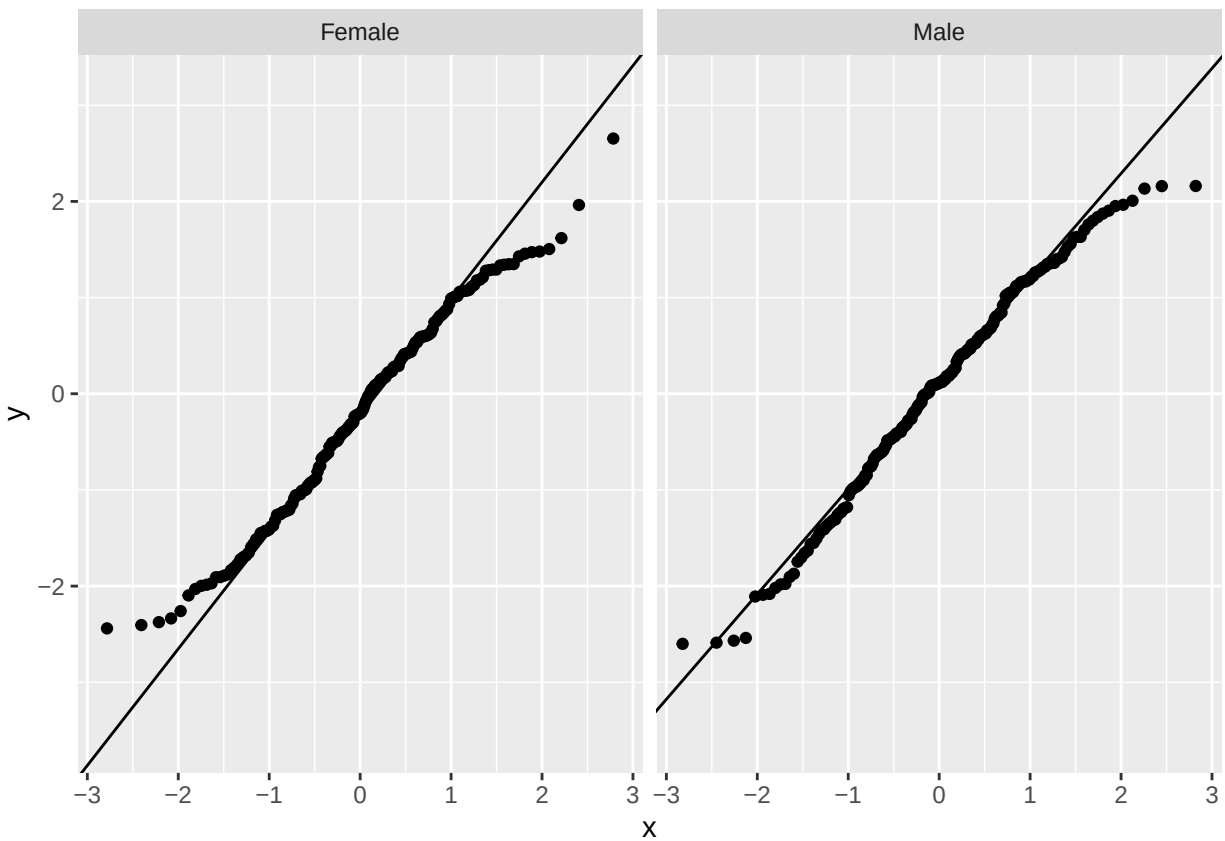
Mracne osobine licnosti po spolu (Male vs Female)



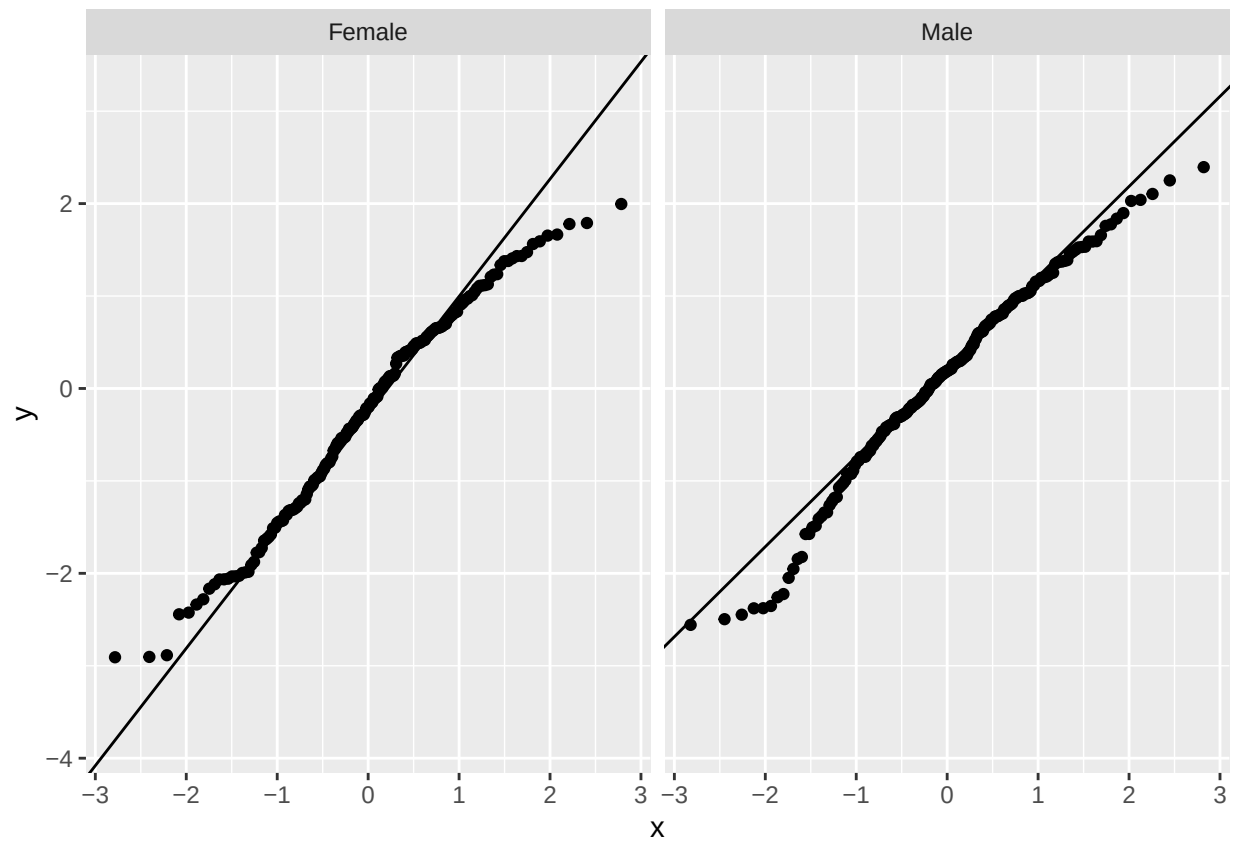
Vizualna analiza mračnih osobina ličnosti ukazuje na izraženije spolne razlike u odnosu na Big Five osobine. Muškarci u prosjeku postižu više rezultate u makijavelizmu, narcizmu, psihoticizmu i sadizmu, što je vidljivo kroz više medijane i pomak distribucija prema višim vrijednostima. Razlike su posebno izražene kod sadizma i psihoticizma, dok su kod makijavelizma i narcizma prisutne umjerene razlike uz djelomično preklapanje distribucija.

Također, za mračne osobine ličnosti provest ćemo provjeru pretpostavki normalnosti distribucije i jednakosti varijanci. Analiza se temelji na istim dvjema nezavisnim skupinama ispitanika (Female i Male).

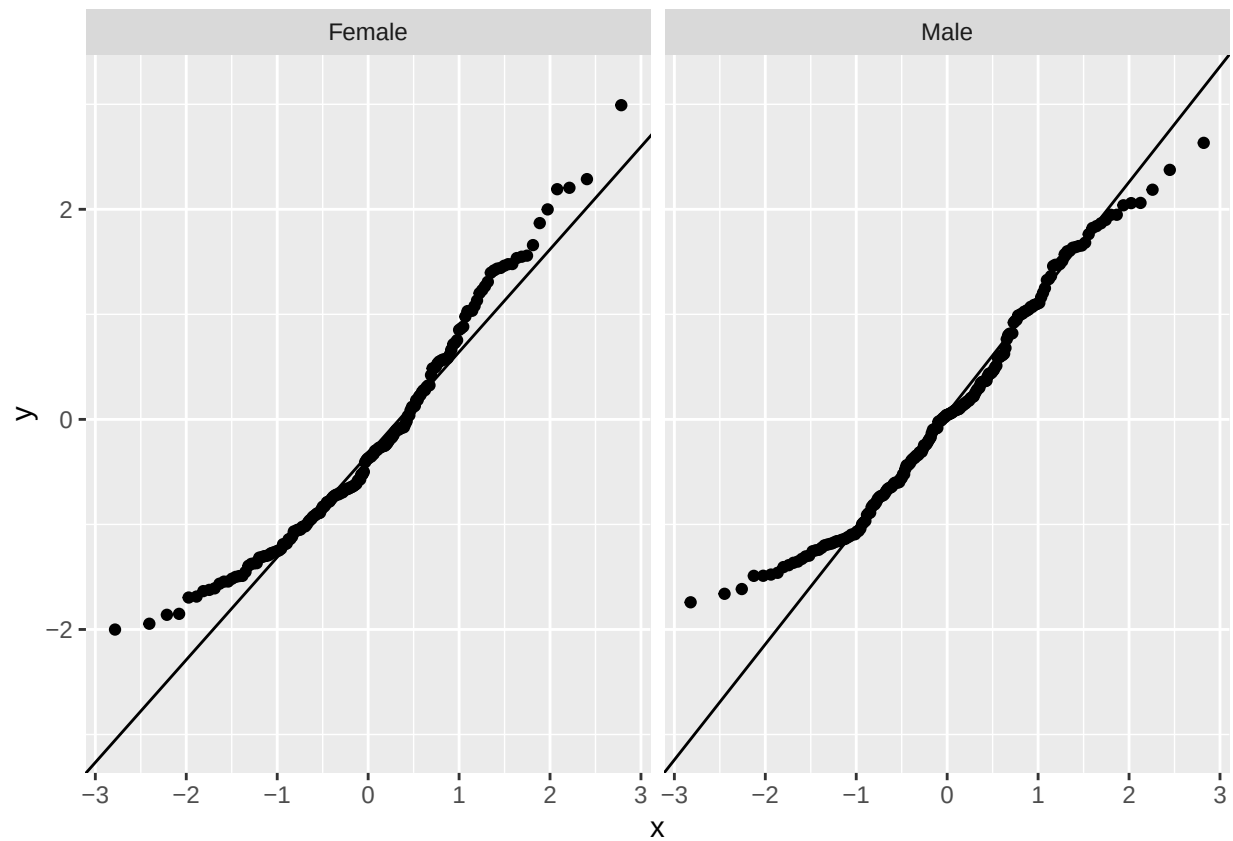
```
ggplot(df_sex, aes(sample = narcissism)) +
  stat_qq() + stat_qq_line() +
  facet_wrap(~sex)
```



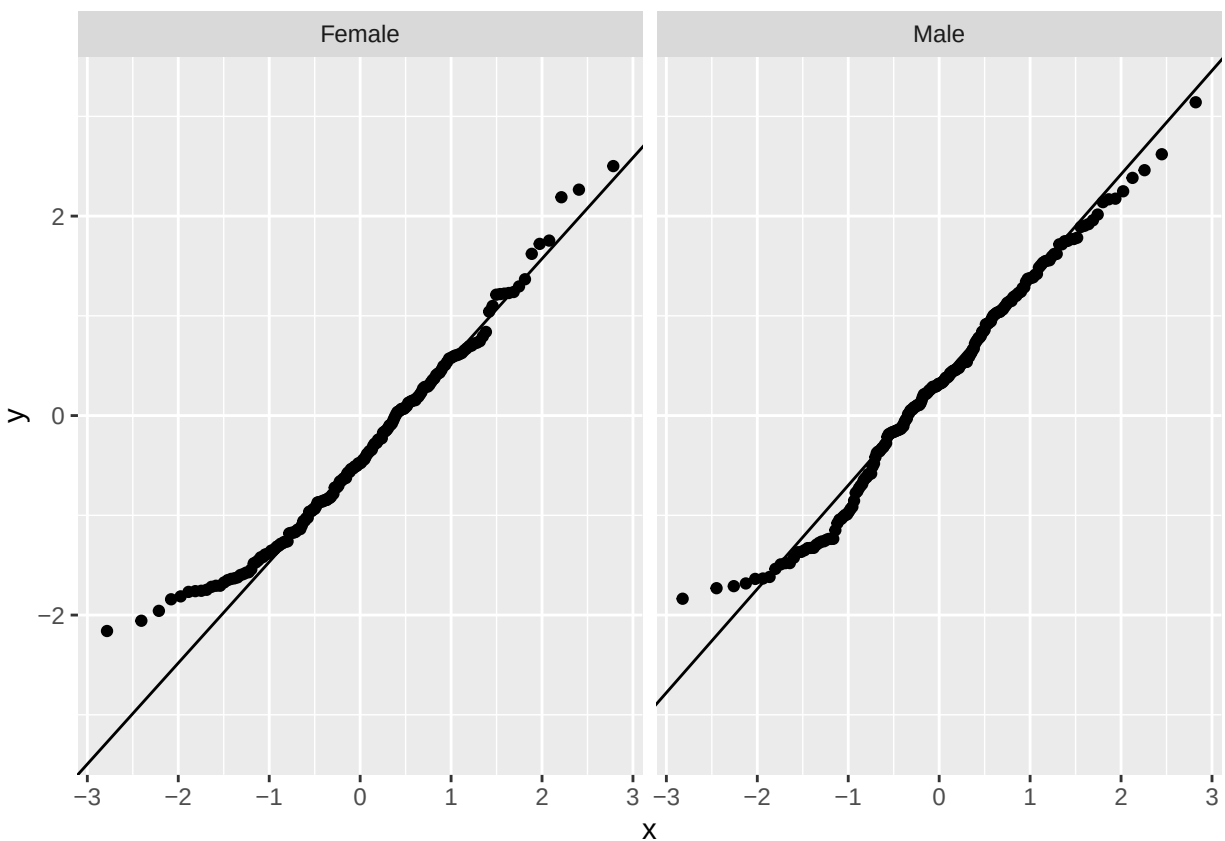
```
ggplot(df_sex, aes(sample = machiavelism)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



```
ggplot(df_sex, aes(sample = psychoticism)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



```
ggplot(df_sex, aes(sample = sadism)) +  
  stat_qq() + stat_qq_line() +  
  facet_wrap(~sex)
```



Distribucije u skupinama su priližno normalne.

```
dark <- c("narcissism", "machiavelism", "psychoticism", "sadism")
```

```
var_by_sex <- df_sex %>%
  group_by(sex) %>%
  summarise(across(all_of(dark), ~ var(.x, na.rm = TRUE)))
```

```
var_by_sex
```

```
## # A tibble: 2 x 5
##   sex    narcissism machiavelism psychoticism sadism
##   <fct>      <dbl>         <dbl>         <dbl>    <dbl>
## 1 Female      1.12           1.24           0.981    0.870
## 2 Male       1.17           1.05           0.967    1.08
```

```
var_tests <- lapply(dark, function(tr) {
  f <- as.formula(paste(tr, "~ sex"))
  vt <- var.test(f, data = df_sex)
  tibble(
    trait = tr,
    F_statistic = round(vt$statistic, 3),
    p_value = round(vt$p.value, 4)
  )
})
```

```
bind_rows(var_tests)
```

```
## # A tibble: 4 x 3
##   trait          F_statistic p_value
##   <chr>          <dbl>   <dbl>
## 1 narcissism      0.957   0.764
## 2 machiavelism    1.18    0.243
## 3 psychoticism    1.01    0.916
## 4 sadism          0.802   0.125
```

Varijance mračnih osobina ličnosti između muškaraca i žena su usporedive i približno jednake, stoga je pretpostavka homogenosti varijanci zadovoljena.

Provest ćemo t-test.

```
dark <- c("narcissism", "machiavelism", "psychoticism", "sadism")

dark_results <- lapply(dark, function(tr) {
  formula <- as.formula(paste(tr, "~ sex"))
  test <- t.test(formula, data = df_sex)
  tibble(
    trait = tr,
    t_statistic = round(test$statistic, 3),
    p_value = round(test$p.value, 5),
    mean_female = round(test$estimate["mean in group Female"], 3),
    mean_male = round(test$estimate["mean in group Male"], 3)
  )
})

dark_results_df <- bind_rows(dark_results)
dark_results_df
```

```
## # A tibble: 4 x 5
##   trait          t_statistic p_value mean_female mean_male
##   <chr>          <dbl>   <dbl>   <dbl>   <dbl>
## 1 narcissism    -2.58 0.0102    -0.221   0.057
## 2 machiavelism  -3.75 0.0002    -0.27    0.136
## 3 psychoticism  -3.22 0.00138   -0.251   0.069
## 4 sadism       -6.80 0          -0.385   0.29
```

U analizi mračnih osobina ličnosti pronađene su jasne spolne razlike. Budući da je u t-testu korišten format `tr ~ sex`, negativne t-vrijednosti ukazuju na više rezultate u skupini muškaraca u odnosu na žene.

Rezultati pokazuju da muškarci postižu statistički značajno više rezultate na svim analiziranim mračnim crtama — narcizmu, makijavelizmu, psihoticizmu i sadizmu (sve $p < 0.05$). Najizraženija razlika uočena je kod sadizma, gdje muškarci ostvaruju znatno više prosječne vrijednosti u odnosu na žene, što upućuje na izraženije spolne razlike u ovoj dimenziji ličnosti.

```
df_sex$sex <- factor(df_sex$sex, levels = c("Female", "Male"))
levels(df_sex$sex)
```

```
## [1] "Female" "Male"
```

```

traits_dark <- c("narcissism", "machiavelism",
                "psychoticism", "sadism")

results_one_tailed <- lapply(traits_dark, function(tr) {
  f <- as.formula(paste(tr, "~ sex"))
  test <- t.test(f, data = df_sex, alternative = "less")

  tibble(
    trait = tr,
    t_statistic = round(test$statistic, 3),
    p_value = round(test$p.value, 5),
    mean_female = round(test$estimate["mean in group Female"], 3),
    mean_male = round(test$estimate["mean in group Male"], 3)
  )
})

results_one_tailed_df <- bind_rows(results_one_tailed)
results_one_tailed_df

```

```

## # A tibble: 4 x 5
##   trait          t_statistic p_value mean_female mean_male
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 narcissism      -2.58 0.0051      -0.221     0.057
## 2 machiavelism    -3.75 0.0001      -0.27      0.136
## 3 psychoticism    -3.22 0.00069     -0.251     0.069
## 4 sadism          -6.80 0          -0.385     0.29

```

Rezultati jednostranog t-testa, provedenog u smjeru teorijske pretpostavke da muškarci postižu više rezultate od žena, potvrđuju postojanje statistički značajnih spolnih razlika u svim analiziranim mračnim osobinama ličnosti. Muškarci ostvaruju značajno više vrijednosti u narcizmu, makijavelizmu, psihoticizmu i sadizmu, pri čemu je razlika najizraženija kod sadizma.

Jačaju li neke osobine s godinama?

Nezavisna varijabla (prediktori):

→ Dob (age)

Zavisne varijable:

→ Ekstraverzija (extraversion) → Ugodnost (agreeableness) → Savjesnost (conscientiousness) → Neuroticizam (neuroticism) → Otvorenost (openness)

U nastavku se analiza provodi zasebno za svaku osobinu. Budući da su i dob i crte ličnosti kontinuirane varijable, a istraživačko pitanje se odnosi na linearnu promjenu s godinama, jednostavna linearna regresija predstavlja najprikladniji statistički pristup.

```

describe(data_clean %>%
  select(age, extraversion, agreeableness,
          conscientiousness, neuroticism, openness))

```

```

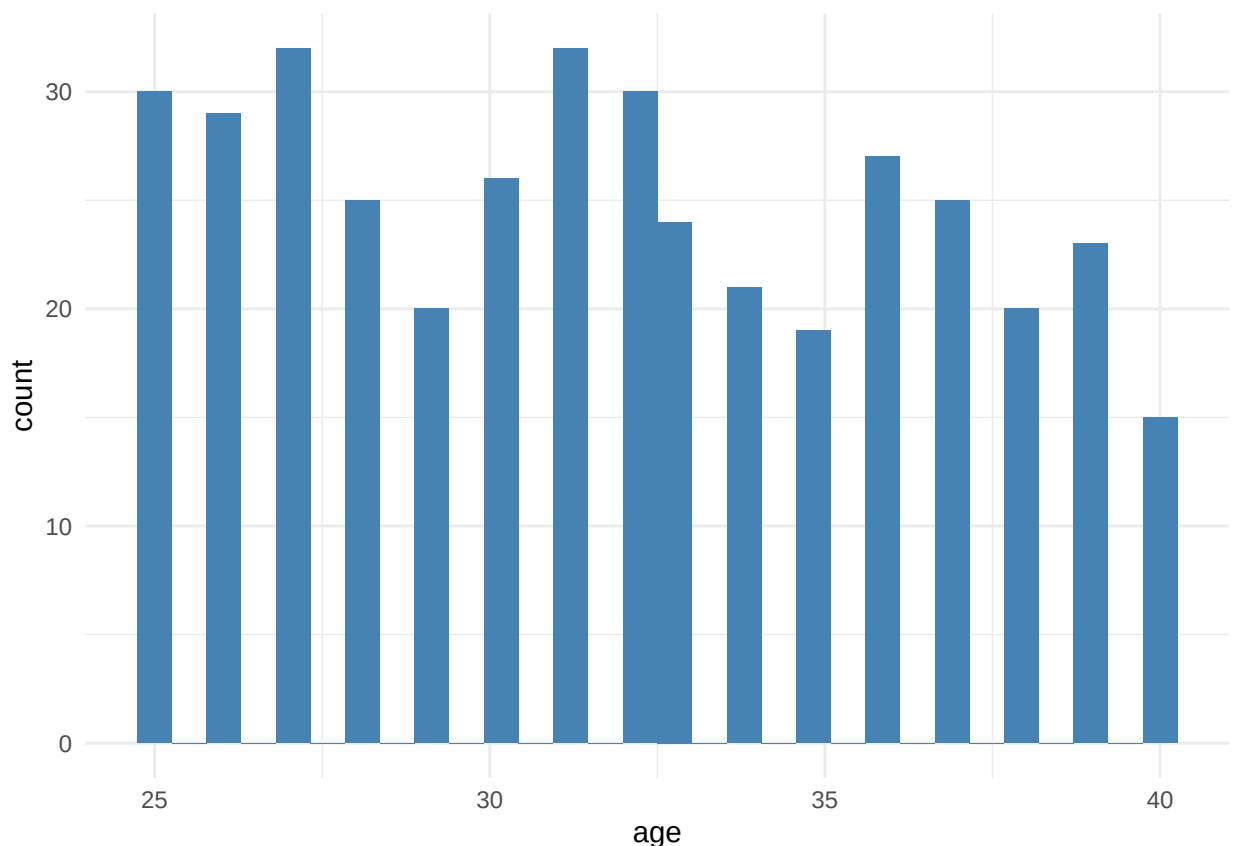
##           vars    n mean   sd median trimmed  mad   min   max range
## age           1 398 31.92 4.54  32.00   31.83 5.93 25.00 40.00 15.00

```


## extraversion	2	398	-0.04	1.09	0.09	0.02	1.05	-3.31	2.54	5.85
## agreeableness	3	398	0.07	1.03	0.10	0.07	1.12	-3.12	2.77	5.89
## conscientiousness	4	398	0.05	0.99	0.07	0.08	1.05	-2.94	2.40	5.34
## neuroticism	5	398	0.05	1.05	-0.07	0.00	1.05	-2.27	2.95	5.22
## openness	6	398	0.04	1.05	-0.09	0.04	0.95	-3.38	2.99	6.36
##			skew	kurtosis	se					
## age			0.12	-1.18	0.23					
## extraversion			-0.50	-0.11	0.05					
## agreeableness			-0.06	-0.33	0.05					
## conscientiousness			-0.26	-0.28	0.05					
## neuroticism			0.41	-0.05	0.05					
## openness			-0.03	0.33	0.05					

Deskriptivna statistika pruža osnovni uvid u raspodjelu dobi i razinu pojedinih crta ličnosti u uzorku ispitanika.

```
ggplot(data_clean, aes(age)) +
  geom_histogram(bins = 30, fill = "steelblue") +
  theme_minimal()
```

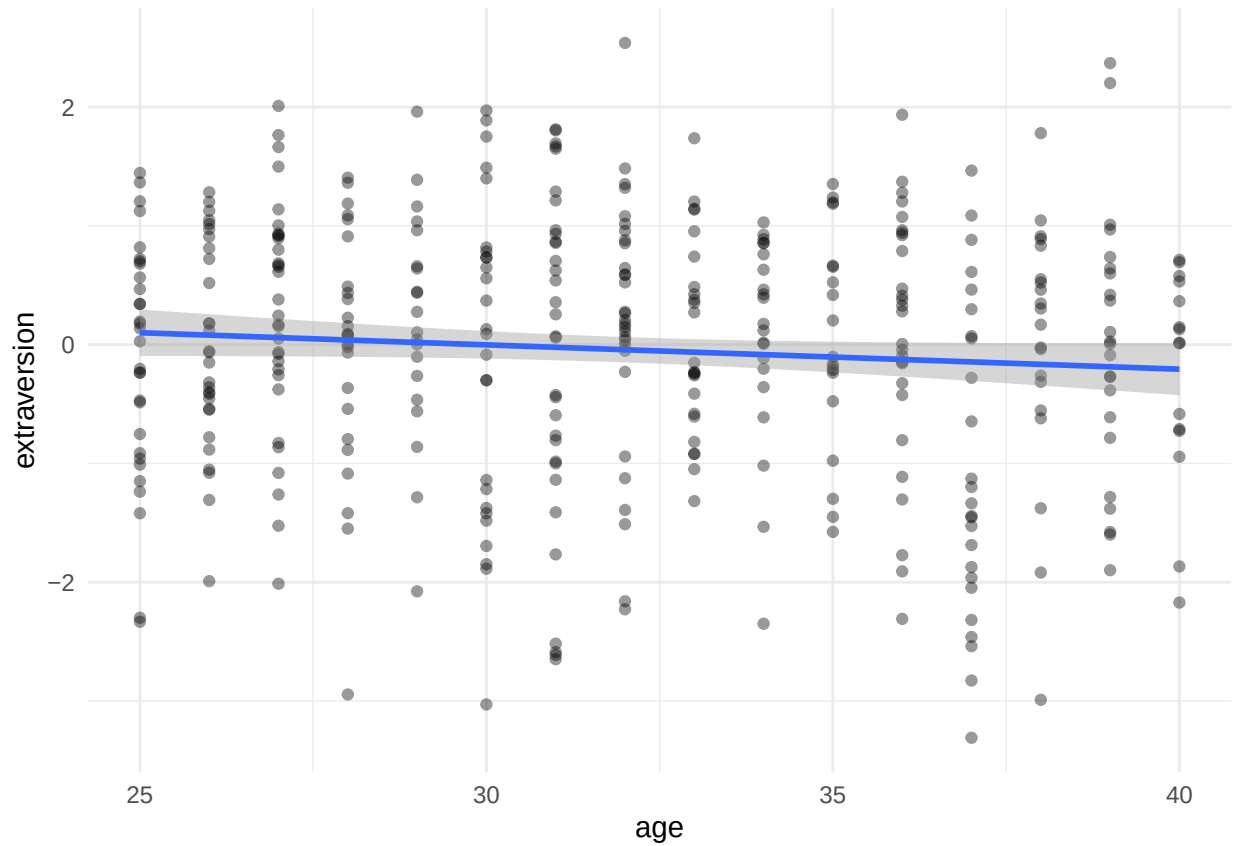


Histogram dobi prikazuje raspodjelu ispitanika po dobnim skupinama te omogućuje procjenu raspona i strukture dobi u uzorku.

```
ggplot(data_clean, aes(age, extraversion)) +
  geom_point(alpha = 0.4) +
```

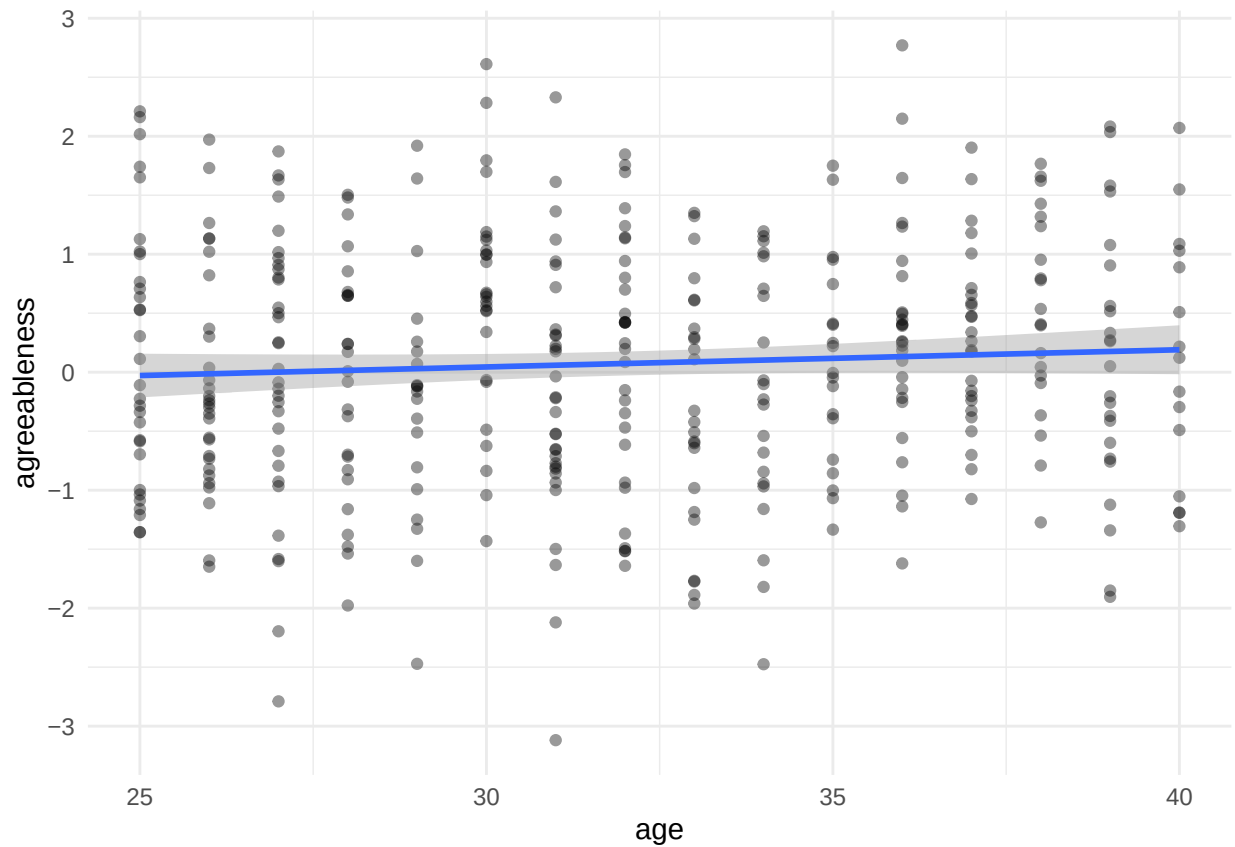
```
geom_smooth(method = "lm", se = TRUE) +  
theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



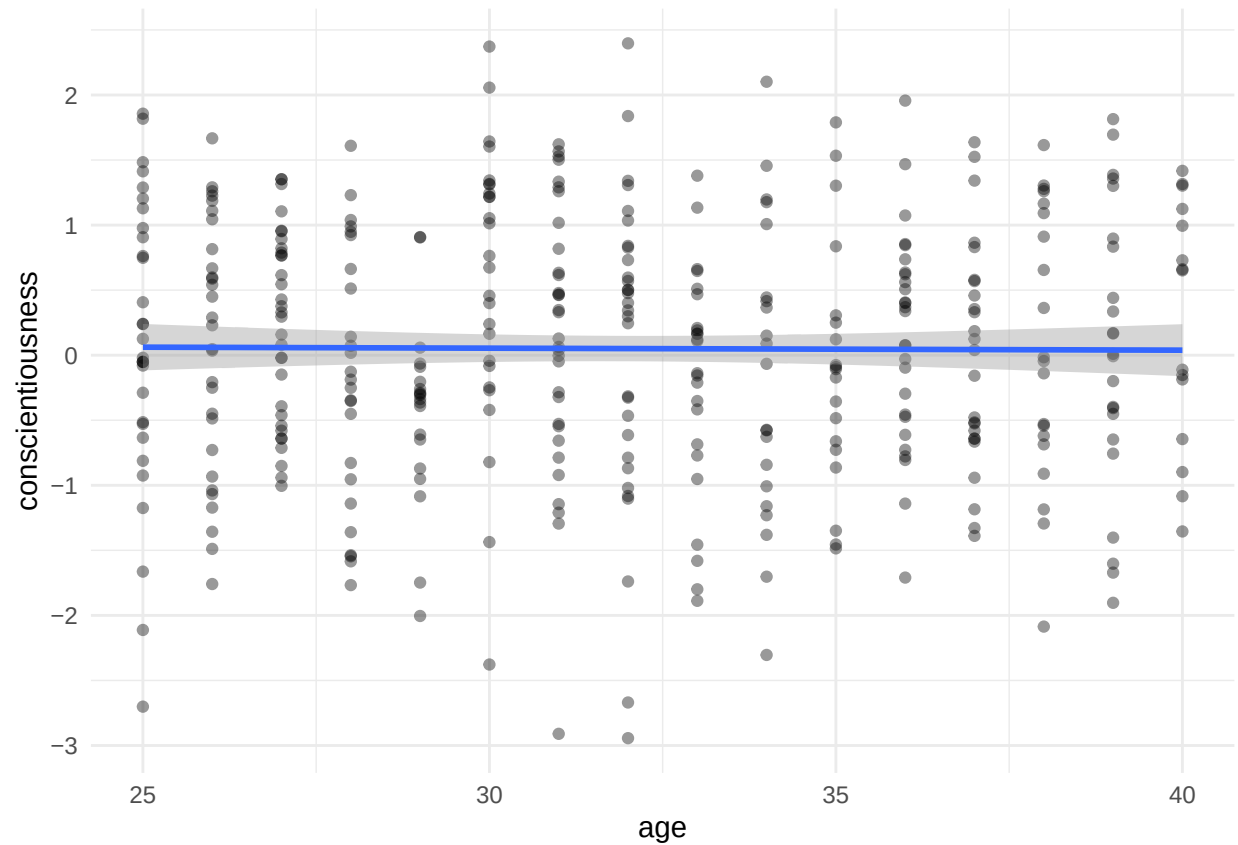
```
ggplot(data_clean, aes(age, agreeableness)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = TRUE) +  
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



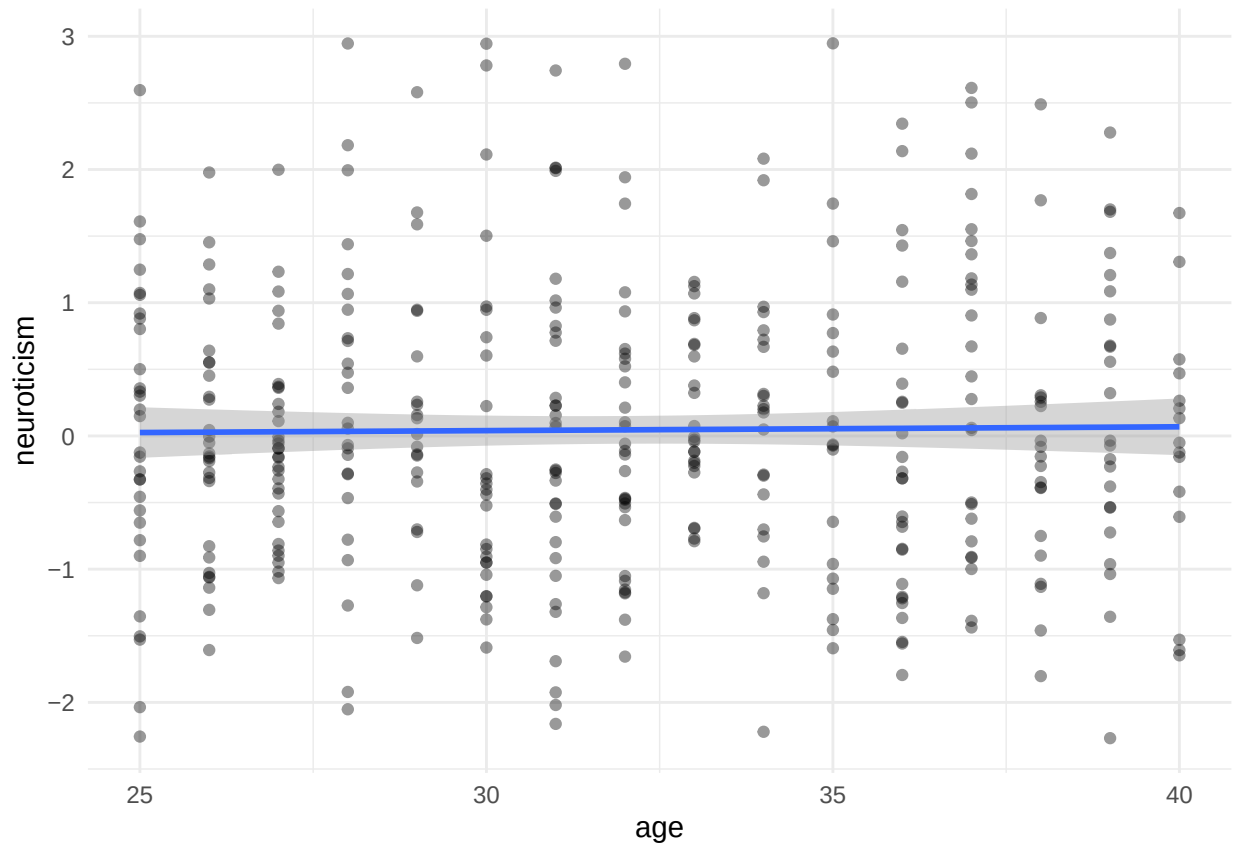
```
ggplot(data_clean, aes(age, conscientiousness)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = TRUE) +  
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



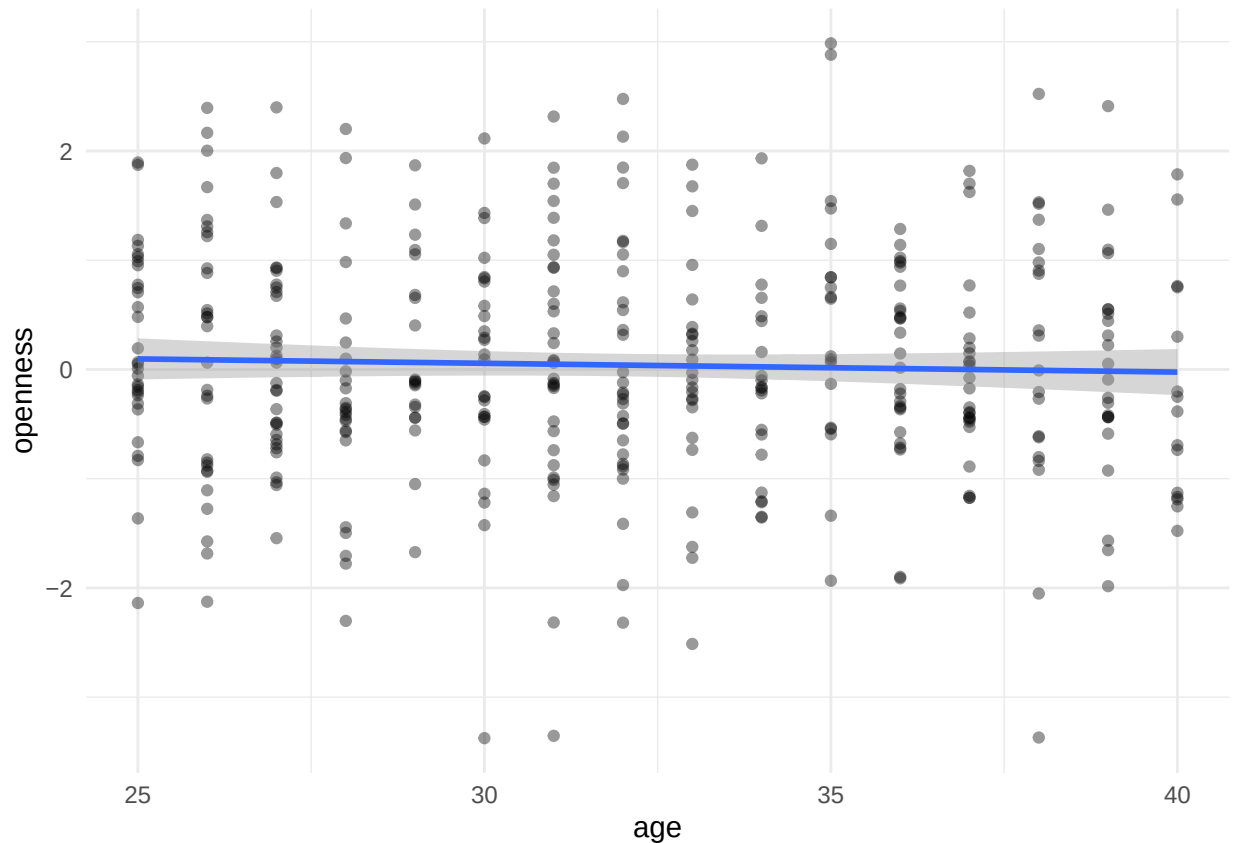
```
ggplot(data_clean, aes(age, neuroticism)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = TRUE) +  
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggplot(data_clean, aes(age, openness)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = TRUE) +  
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Raspršeni dijagrami s pripadajućim regresijskim linijama korišteni su za vizualnu procjenu linearnog odnosa između dobi i pojedinih crta ličnosti. Grafički prikazi ne upućuju na izražene linearne trendove, iako se kod nekih osobina može uočiti blagi rast ili pad vrijednosti s porastom dobi.

```
cor.test(data_clean$age, data_clean$extraversion)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$age and data_clean$extraversion
## t = -1.7053, df = 396, p-value = 0.08893
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.18215014 0.01302653
## sample estimates:
## cor
## -0.08538084
```

```
cor.test(data_clean$age, data_clean$agreeableness)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$age and data_clean$agreeableness
## t = 1.2789, df = 396, p-value = 0.2017
```

```
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.0343782  0.1614170
## sample estimates:
##      cor
## 0.06413658
```

```
cor.test(data_clean$age, data_clean$conscientiousness)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$age and data_clean$conscientiousness
## t = -0.13972, df = 396, p-value = 0.889
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.10524642  0.09134006
## sample estimates:
##      cor
## -0.007021018
```

```
cor.test(data_clean$age, data_clean$neuroticism)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$age and data_clean$neuroticism
## t = 0.2454, df = 396, p-value = 0.8063
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.08607148  0.11049500
## sample estimates:
##      cor
## 0.01233089
```

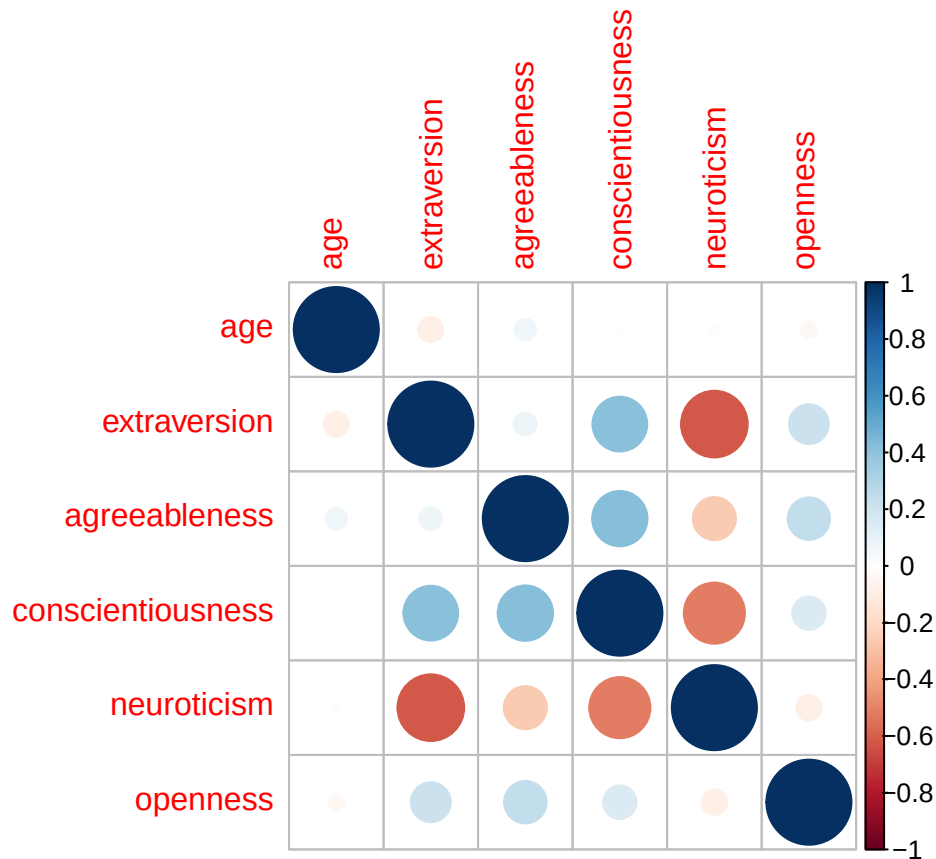
```
cor.test(data_clean$age, data_clean$openness)
```

```
##
## Pearson's product-moment correlation
##
## data: data_clean$age and data_clean$openness
## t = -0.69464, df = 396, p-value = 0.4877
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.13272862  0.06363051
## sample estimates:
##      cor
## -0.03488574
```

Pearsonove korelacije korištene su kao preliminarna analiza odnosa dobi i crta ličnosti. Rezultati korelacija ne upućuju na statistički značajne povezanosti između dobi i promatranih osobina.

```
vars <- data_clean %>%
  select(age, extraversion, agreeableness,
         conscientiousness, neuroticism, openness) %>%
  na.omit()

corrplot(cor(vars))
```



Korelacijska matrica dodatno potvrđuje izostanak snažnih povezanosti između dobi i pojedinih crta ličnosti.

```
model_consc <- lm(conscientiousness ~ age, data = data_clean)
summary(model_consc)
```

```
##
## Call:
## lm(formula = conscientiousness ~ age, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99434 -0.68117  0.01346  0.76485  2.34600
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.099902   0.353798   0.282   0.778
## age         -0.001533   0.010973  -0.140   0.889
##
```



```
## Residual standard error: 0.9919 on 396 degrees of freedom
## Multiple R-squared:  4.929e-05, Adjusted R-squared:  -0.002476
## F-statistic: 0.01952 on 1 and 396 DF,  p-value: 0.889
```

```
model_extra <- lm(extraversion ~ age, data = data_clean)
summary(model_extra)
```

```
##
## Call:
## lm(formula = extraversion ~ age, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1645 -0.6666  0.1027  0.7857  2.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.61250     0.38726   1.582  0.1145
## age        -0.02048     0.01201  -1.705  0.0889 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.086 on 396 degrees of freedom
## Multiple R-squared:  0.00729, Adjusted R-squared:  0.004783
## F-statistic: 2.908 on 1 and 396 DF,  p-value: 0.08893
```

```
model_agree <- lm(agreeableness ~ age, data = data_clean)
summary(model_agree)
```

```
##
## Call:
## lm(formula = agreeableness ~ age, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1780 -0.7163  0.0236  0.7345  2.6385
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.39279     0.36742  -1.069  0.286
## age          0.01457     0.01140   1.279  0.202
##
## Residual standard error: 1.03 on 396 degrees of freedom
## Multiple R-squared:  0.004114, Adjusted R-squared:  0.001599
## F-statistic: 1.636 on 1 and 396 DF,  p-value: 0.2017
```

```
model_neuro <- lm(neuroticism ~ age, data = data_clean)
summary(model_neuro)
```

```
##
## Call:
## lm(formula = neuroticism ~ age, data = data_clean)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3342 -0.7428 -0.1191  0.6403  2.9115
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.045980   0.376568  -0.122   0.903
## age          0.002866   0.011679   0.245   0.806
##
## Residual standard error: 1.056 on 396 degrees of freedom
## Multiple R-squared:  0.0001521, Adjusted R-squared:  -0.002373
## F-statistic: 0.06022 on 1 and 396 DF,  p-value: 0.8063
```

```
model_open <- lm(openness ~ age, data = data_clean)
summary(model_open)
```

```
##
## Call:
## lm(formula = openness ~ age, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4319 -0.6082 -0.1301  0.6922  2.9697
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.297455   0.373452   0.796   0.426
## age         -0.008046   0.011583  -0.695   0.488
##
## Residual standard error: 1.047 on 396 degrees of freedom
## Multiple R-squared:  0.001217, Adjusted R-squared:  -0.001305
## F-statistic: 0.4825 on 1 and 396 DF,  p-value: 0.4877
```

Za svaku crtu ličnosti provedena je analiza jednostavne linearne regresije u kojoj je dob korištena kao nezavisna varijabla. U niti jednom modelu dob se nije pokazala statistički značajnim prediktorom ($p > 0.05$). Vrijednosti koeficijenta determinacije bile su vrlo niske, što upućuje na slab objašnjavajući doprinos dobi.

Na temelju provedene analize može se zaključiti da se u promatranom uzorku ne uočava statistički značajno jačanje niti slabljenje crta ličnosti s godinama. Dob se u ovom istraživanju nije pokazala značajnim prediktorom promjena u okviru Big Five modela ličnosti.

```
library(broom)

models <- list(
  Extraversion = model_extra,
  Agreeableness = model_agree,
  Conscientiousness = model_consc,
  Neuroticism = model_neuro,
  Openness = model_open
)

results <- lapply(models, tidy) |>
```

```
bind_rows(.id = "Trait") |>
  filter(term == "age")
```

results

```
## # A tibble: 5 x 6
##   Trait      term estimate std.error statistic p.value
##   <chr>      <chr>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Extraversion age    -0.0205    0.0120    -1.71    0.0889
## 2 Agreeableness age     0.0146    0.0114     1.28    0.202
## 3 Conscientiousness age   -0.00153    0.0110    -0.140   0.889
## 4 Neuroticism age     0.00287    0.0117     0.245   0.806
## 5 Openness age    -0.00805    0.0116    -0.695   0.488
```

Tablica prikazuje procijenjene regresijske koeficijente za učinak dobi na pojedine crte ličnosti.

Kategorizacija kontinuiranih varijabli provedena je radi primjene hi-kvadrat testa, kojim se ispituje povezanost između dobnih skupina i distribucije razina crta ličnosti.

```
data_clean <- data_clean %>%
  mutate(age_group = case_when(
    age < 30 ~ "Mladi",
    age >= 30 & age < 35 ~ "Srednji",
    age >= 35 ~ "Stariji"
  ))

data_clean$age_group <- factor(data_clean$age_group)
```

Crte ličnosti podijeljene su u tri kategorije (niska, srednja i visoka razina) na temelju tercila distribucije rezultata.

```
traits <- c("extraversion", "agreeableness", "conscientiousness", "neuroticism", "openness")

for (t in traits) {
  data_clean[[paste0(t, "_cat")]] <- cut(
    data_clean[[t]],
    breaks = quantile(data_clean[[t]], probs = c(0, 1/3, 2/3, 1), na.rm = TRUE),
    labels = c("Niska", "Srednja", "Visoka"),
    include.lowest = TRUE
  )
}
```

```
chisq.test(table(data_clean$age_group, data_clean$extraversion_cat))
```

```
##
## Pearson's Chi-squared test
##
## data:  table(data_clean$age_group, data_clean$extraversion_cat)
## X-squared = 2.7214, df = 4, p-value = 0.6055
```

```
chisq.test(table(data_clean$age_group, data_clean$agreeableness_cat))
```

```
##  
## Pearson's Chi-squared test  
##  
## data: table(data_clean$age_group, data_clean$agreeableness_cat)  
## X-squared = 8.7962, df = 4, p-value = 0.0664
```

```
chisq.test(table(data_clean$age_group, data_clean$conscientiousness_cat))
```

```
##  
## Pearson's Chi-squared test  
##  
## data: table(data_clean$age_group, data_clean$conscientiousness_cat)  
## X-squared = 1.176, df = 4, p-value = 0.882
```

```
chisq.test(table(data_clean$age_group, data_clean$neuroticism_cat))
```

```
##  
## Pearson's Chi-squared test  
##  
## data: table(data_clean$age_group, data_clean$neuroticism_cat)  
## X-squared = 5.0875, df = 4, p-value = 0.2784
```

```
chisq.test(table(data_clean$age_group, data_clean$openness_cat))
```

```
##  
## Pearson's Chi-squared test  
##  
## data: table(data_clean$age_group, data_clean$openness_cat)  
## X-squared = 0.79106, df = 4, p-value = 0.9396
```

```
table(data_clean$age_group, data_clean$openness_cat)
```

```
##  
##           Niska Srednja Visoka  
## Mladi      44      44      48  
## Srednji    45      47      41  
## Stariji    44      41      44
```

Hi-kvadrat testom ispitana je povezanost između dobnih skupina i kategoriziranih razina crta ličnosti. Rezultati nisu pokazali statistički značajnu povezanost između dobi i ekstraverzije, savjesnosti, neuroticizma niti otvorenosti ($p > 0.05$).

Rezultati hi-kvadrat testova nisu pokazali statistički značajnu povezanost između dobnih skupina i osobina.

Kod osobine ugodnosti uočen je povišen hi-kvadrat rezultat (8.80), no pripadajuća p-vrijednost (0.066) ne doseže prag statističke značajnosti. Stoga se ne može zaključiti da se razine ugodnosti značajno razlikuju između dobnih skupina.

Iako je kod ugodnosti uočena blaga tendencija promjene raspodjele razina između dobnih skupina, ona nije statistički potvrđena, što može biti posljedica kategorizacije kontinuiranih varijabli i smanjene statističke snage hi-kvadrat testa.

Na temelju hi-kvadrat analize kategoriziranih varijabli ne može se zaključiti da se crte ličnosti značajno mijenjaju ili jačaju s godinama u promatranom uzorku.

Kako bi se istovremeno procijenio doprinos više prediktora u objašnjenju varijabilnosti crta ličnosti, primijenjena je višestruka linearna regresija u kojoj su dob i ostale relevantne varijable uključene kao nezavisni prediktori, dok su crte ličnosti tretirane kao kontinuirane zavisne varijable.

```
data_clean$occupation <- factor(data_clean$"employment status")
```

```
data_clean$occupation <- fct_lump_min(  
  data_clean$occupation,  
  min = 10  
)
```

```
chisq.test(  
  table(data_clean$occupation, data_clean$extraversion_cat),simulate.p.value = TRUE  
)
```

```
##  
## Pearson's Chi-squared test with simulated p-value (based on 2000  
## replicates)  
##  
## data:  table(data_clean$occupation, data_clean$extraversion_cat)  
## X-squared = 33.966, df = NA, p-value = 0.0004998
```

```
chisq.test(  
  table(data_clean$occupation, data_clean$agreeableness_cat),simulate.p.value = TRUE  
)
```

```
##  
## Pearson's Chi-squared test with simulated p-value (based on 2000  
## replicates)  
##  
## data:  table(data_clean$occupation, data_clean$agreeableness_cat)  
## X-squared = 6.2437, df = NA, p-value = 0.6247
```

```
chisq.test(  
  table(data_clean$occupation, data_clean$conscientiousness_cat),simulate.p.value = TRUE  
)
```

```
##  
## Pearson's Chi-squared test with simulated p-value (based on 2000  
## replicates)  
##  
## data:  table(data_clean$occupation, data_clean$conscientiousness_cat)  
## X-squared = 20.033, df = NA, p-value = 0.009495
```

```
chisq.test(
  table(data_clean$occupation, data_clean$neuroticism_cat),simulate.p.value = TRUE
)
```

```
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data:  table(data_clean$occupation, data_clean$neuroticism_cat)
## X-squared = 35.602, df = NA, p-value = 0.0004998
```

```
chisq.test(
  table(data_clean$occupation, data_clean$openness_cat),simulate.p.value = TRUE
)
```

```
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data:  table(data_clean$occupation, data_clean$openness_cat)
## X-squared = 10.37, df = NA, p-value = 0.2489
```

Zbog niskih očekivanih frekvencija u pojedinim ćelijama korištena je Monte Carlo simulacija p-vrijednosti.

Ovi nalazi upućuju na to da se razine pojedinih crta ličnosti razlikuju između skupina različitog zanimanja. Zanimanje se pokazalo značajno povezanim s ekstraverzijom, savjesnošću i neuroticizmom, dok za ugodnost i otvorenost takva povezanost nije utvrđena. Time se sugerira da socijalni i radni kontekst može biti povezan s izraženošću određenih crta ličnosti.

```
library(rcompanion)
```

```
##
## Attaching package: 'rcompanion'

## The following object is masked from 'package:psych':
##
##      phi
```

```
# Ekstraverzija
cramerV(table(data_clean$occupation, data_clean$extraversion_cat))
```

```
## Cramer V
##      0.2066
```

```
prop_extr <- prop.table(
  table(data_clean$occupation, data_clean$extraversion_cat),
  1
)

round(prop_extr * 100, 1)
```

```
##
##
##      Full-Time      Niska Srednja Visoka
##      Not in paid work (e.g. homemaker', 'retired or disabled) 73.3 20.0 6.7
##      Part-Time      42.9 23.2 33.9
##      Unemployed (and job seeking) 57.1 25.0 17.9
##      Other          52.4 33.3 14.3
```

```
# Savjesnost
```

```
cramerV(table(data_clean$occupation, data_clean$conscientiousness_cat))
```

```
## Cramer V
```

```
## 0.1586
```

```
prop_consc <- prop.table(
  table(data_clean$occupation, data_clean$conscientiousness_cat),
  1
)
```

```
round(prop_consc * 100, 1)
```

```
##
##
##      Full-Time      Niska Srednja Visoka
##      Not in paid work (e.g. homemaker', 'retired or disabled) 30.9 32.4 36.7
##      Part-Time      33.3 53.3 13.3
##      Unemployed (and job seeking) 28.6 33.9 37.5
##      Other          64.3 21.4 14.3
##      Other          38.1 42.9 19.0
```

```
# Neuroticizam
```

```
cramerV(table(data_clean$occupation, data_clean$neuroticism_cat))
```

```
## Cramer V
```

```
## 0.2115
```

```
prop_neuro <- prop.table(
  table(data_clean$occupation, data_clean$neuroticism_cat),
  1
)
```

```
round(prop_neuro * 100, 1)
```

```
##
##
##      Full-Time      Niska Srednja Visoka
##      Not in paid work (e.g. homemaker', 'retired or disabled) 38.1 33.8 28.1
##      Part-Time      6.7 13.3 80.0
##      Unemployed (and job seeking) 32.1 41.1 26.8
##      Other          14.3 21.4 64.3
##      Other          19.0 33.3 47.6
```

Dok analize prema dobnim skupinama nisu pokazale statistički značajne razlike u crtama ličnosti, rezultati prema zanimanju sugeriraju da se ekstraverzija, savjesnost i neuroticizam mogu razlikovati ovisno o socijalnom/profesionalnom kontekstu, a ne nužno o kronološkoj dobi.

Ukupno gledano, zanimanje je statistički značajno povezano s ekstraverzijom, savjesnošću i neuroticizmom, pri čemu su efekti slabi do umjereni; pritom se u skupinama nezaposlenih i osoba izvan plaćenog rada češće uočavaju nepovoljniji profili (niža savjesnost i viši neuroticizam), dok su zaposleni s punim radnim vremenom češće u niskom/srednjem neuroticizmu te srednjoj/visokoj savjesnosti i ekstraverziji.

Budući da su skupine zanimanja različitih veličina, zaključci se temelje na relativnim razlikama u raspodjelama (udjelima) kategorija unutar skupina, a ne na apsolutnim frekvencijama.

```
make_percent_table <- function(x, row_var, col_var) {
  tab <- prop.table(table(x[[row_var]], x[[col_var]]), 1) * 100
  df <- as.data.frame.matrix(round(tab, 1))
  df[[row_var]] <- rownames(df)
  rownames(df) <- NULL
  df <- df[, c(row_var, colnames(df)[colnames(df) != row_var])]
  df
}

df_extr <- make_percent_table(data_clean, "occupation", "extraversion_cat")
df_consc <- make_percent_table(data_clean, "occupation", "conscientiousness_cat")
df_neuro <- make_percent_table(data_clean, "occupation", "neuroticism_cat")

df_extr
```

```
##                                occupation Niska Srednja Visoka
## 1                                Full-Time 25.5    36.7    37.8
## 2 Not in paid work (e.g. homemaker', 'retired or disabled) 73.3    20.0     6.7
## 3                                Part-Time 42.9    23.2    33.9
## 4                                Unemployed (and job seeking) 57.1    25.0    17.9
## 5                                Other 52.4    33.3    14.3
```

```
df_consc
```

```
##                                occupation Niska Srednja Visoka
## 1                                Full-Time 30.9    32.4    36.7
## 2 Not in paid work (e.g. homemaker', 'retired or disabled) 33.3    53.3    13.3
## 3                                Part-Time 28.6    33.9    37.5
## 4                                Unemployed (and job seeking) 64.3    21.4    14.3
## 5                                Other 38.1    42.9    19.0
```

```
df_neuro
```

```
##                                occupation Niska Srednja Visoka
## 1                                Full-Time 38.1    33.8    28.1
## 2 Not in paid work (e.g. homemaker', 'retired or disabled)  6.7    13.3    80.0
## 3                                Part-Time 32.1    41.1    26.8
## 4                                Unemployed (and job seeking) 14.3    21.4    64.3
## 5                                Other 19.0    33.3    47.6
```


Jesu li crte ličnosti i sklonost stresu, depresiji i anksioznosti povezane sa zanimanjem?

```
df_job <- data_clean %>% mutate(
  occupation_group = if_else(
    `student status` == "Yes",
    "Student",
    `employment status`
  )
) %>% filter(!occupation_group %in% c("Other", "Due to start a new job within the next month", "Not in
table(df_job$occupation_group)
```

```
##
##               Full-Time               Part-Time
##               201               36
##               Student Unemployed (and job seeking)
##               109               24
```

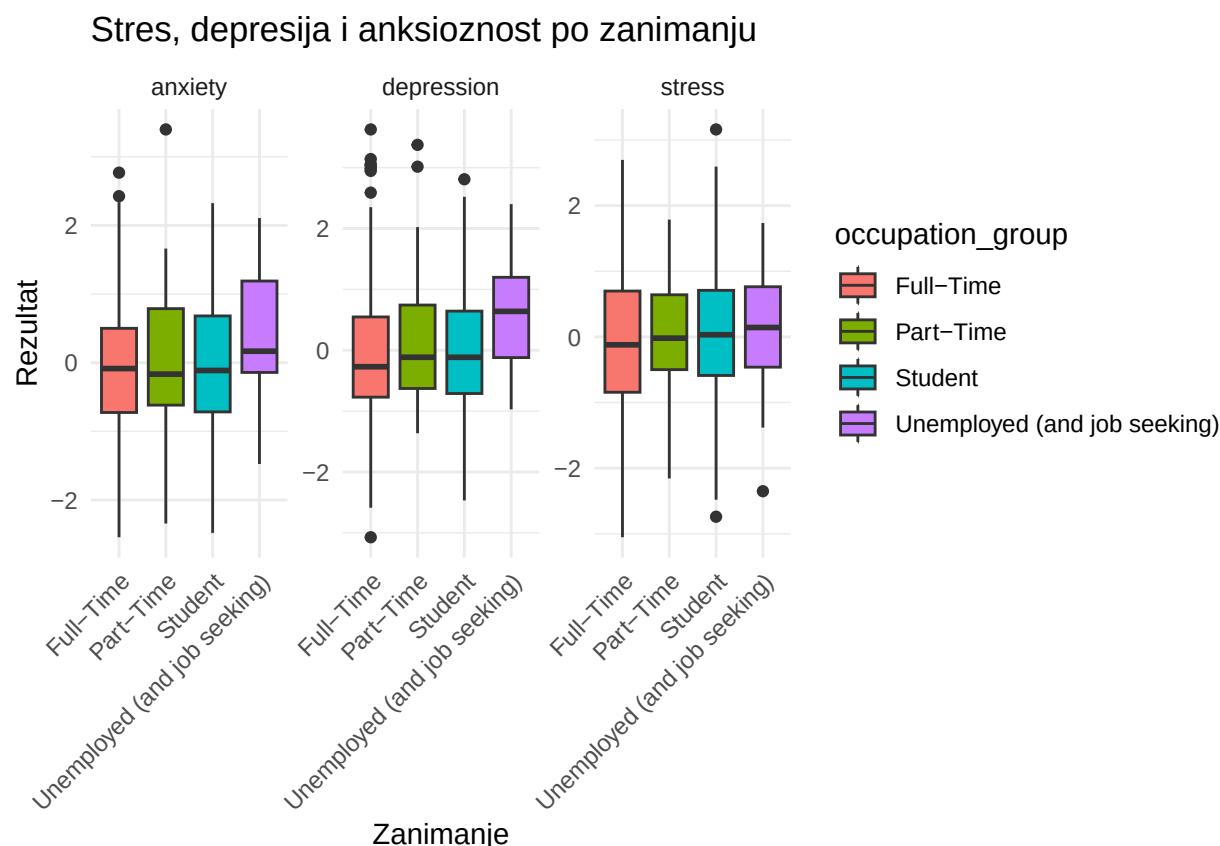
```
df_job %>%
  group_by(occupation_group) %>%
  summarise(
    n = n(),
    mean_stress = mean(stress, na.rm = TRUE),
    mean_depression = mean(depression, na.rm = TRUE),
    mean_anxiety = mean(anxiety, na.rm = TRUE)
  )
```

```
## # A tibble: 4 x 5
##   occupation_group      n mean_stress mean_depression mean_anxiety
##   <chr>             <int>      <dbl>         <dbl>         <dbl>
## 1 Full-Time        201    -0.0589        -0.0448        -0.0169
## 2 Part-Time         36     0.0756         0.127         0.107
## 3 Student          109     0.0739         0.0126        -0.0677
## 4 Unemployed (and job seeking) 24     0.109         0.561         0.346
```

```
mental_long <- df_job %>%
  pivot_longer(
    cols = c(stress, depression, anxiety),
    names_to = "outcome",
    values_to = "score"
  )

ggplot(mental_long, aes(x = occupation_group, y = score, fill = occupation_group)) +
  geom_boxplot() +
  facet_wrap(~ outcome, scales = "free_y") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Stres, depresija i anksioznost po zanimanju",
    x = "Zanimanje",
```

```
y = "Rezultat"
)
```



Vizualni pregled distribucija pokazuje da se razine stresa, depresije i anksioznosti uvelike preklapaju između skupina definiranih prema radnom statusu (zanimanju), što upućuje na izražene individualne razlike unutar skupina. Iako nezaposleni sudionici pokazuju viši medijan depresivnosti u odnosu na ostale skupine, razlike nisu statistički potvrđene. Također je vidljivo da se veći broj ekstremnih vrijednosti javlja u skupini zaposlenih s punim radnim vremenom, što dodatno ukazuje na veliku heterogenost unutar te skupine. Razine stresa pokazuju vrlo slične obrasce u svim skupinama, dok se anksioznost ne pojavljuje sustavno povezanom sa zanimanjem. Ovi nalazi sugeriraju da radni status sam po sebi ima ograničen doprinos objašnjenju mentalnog zdravlja te da individualne crte ličnosti vjerojatno imaju snažniju ulogu.

Provođenje ANOVA analize - Razlikuju li se razine anksioznosti, depresije i stresa po zanimanju (radnim statusima)?

Za početak potrebno je provesti testove pretpostavki normalnosti podataka i homogenosti varijanci za anksioznost, depresiju i stres.

Test pretpostavke normalnosti za anksioznost

```
require(nortest)
```

```
# Lilliefors test normalnosti (ukupno)
lillie.test(df_job$anxiety)
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$anxiety
## D = 0.039549, p-value = 0.1715
```

```
# Lilliefors test po skupinama zanimanja
lillie.test(df_job$anxiety[df_job$occupation_group == "Full-Time"])
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$anxiety[df_job$occupation_group == "Full-Time"]
## D = 0.061511, p-value = 0.06185
```

```
lillie.test(df_job$anxiety[df_job$occupation_group == "Part-Time"])
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$anxiety[df_job$occupation_group == "Part-Time"]
## D = 0.15146, p-value = 0.03593
```

```
lillie.test(df_job$anxiety[df_job$occupation_group == "Unemployed (and job seeking)"])
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$anxiety[df_job$occupation_group == "Unemployed (and job seeking)"]
## D = 0.11293, p-value = 0.5953
```

```
lillie.test(df_job$anxiety[df_job$occupation_group == "Student"])
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$anxiety[df_job$occupation_group == "Student"]
## D = 0.042081, p-value = 0.9066
```

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
```

```
hist(
  df_job$anxiety[df_job$occupation_group == "Full-Time"],
  main = "Anxiety - Full-Time",
  xlab = "Anxiety",
  col = "lightgray",
```

```

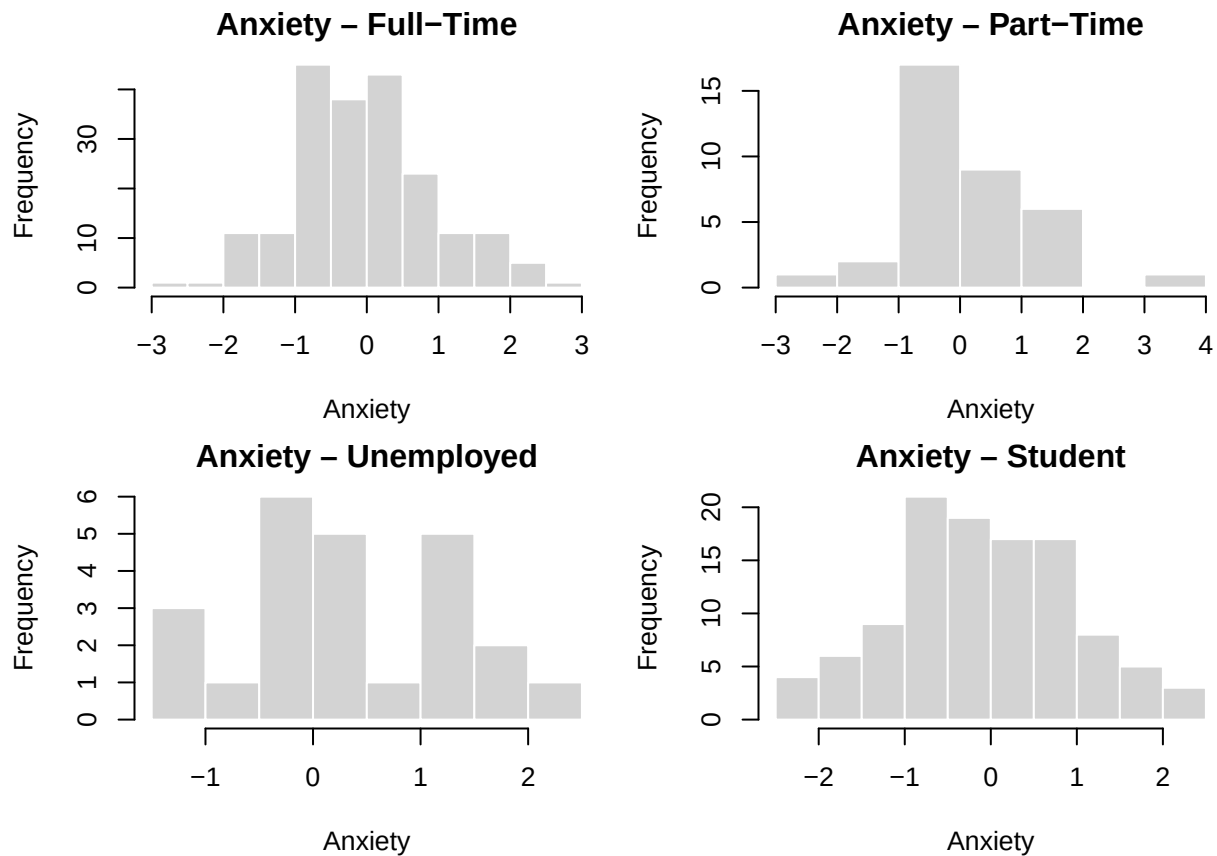
    border = "white"
)

hist(
  df_job$anxiety[df_job$occupation_group == "Part-Time"],
  main = "Anxiety - Part-Time",
  xlab = "Anxiety",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$anxiety[df_job$occupation_group == "Unemployed (and job seeking)"],
  main = "Anxiety - Unemployed",
  xlab = "Anxiety",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$anxiety[df_job$occupation_group == "Student"],
  main = "Anxiety - Student",
  xlab = "Anxiety",
  col = "lightgray",
  border = "white"
)

```



```
par(mfrow = c(1, 1))
```

Iz dobivenih ispisa zaključujemo da na temelju cijelog uzorka nema potrebe za odbacivanjem nulte hipoteze o normalnosti uzorka ($p > 0.05$). Gledajući svaku skupinu radnog statusa zasebno, možemo uočiti kako jedino statistički značajno odstupanje od normalnosti postoji kod ispitanika koji rade na nepuno radno vrijeme, ali to možemo izuzeti iz općeg zaključka jer se ta skupina sastoji od malog uzorka na koje su testovi o normalnosti osjetljivi.

Gledajući histograme, manja odstupanja od simetrije vidljiva su u skupinama s manjim brojem ispitanika (osobito kod zaposlenih s nepunim radnim vremenom i nezaposlenih), što je vjerojatno posljedica veličine uzorka, a ne stvarnih razlika u raspodjeli anksioznosti.

Test pretpostavke normalnosti za depresiju

```
require(nortest)

# Lilliefors test normalnosti (ukupno)
lillie.test(df_job$depression)

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$depression
## D = 0.089732, p-value = 1.443e-07
```

```

# Lilliefors test po skupinama zanimanja
lillie.test(df_job$depression[df_job$occupation_group == "Full-Time"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$depression[df_job$occupation_group == "Full-Time"]
## D = 0.10029, p-value = 4.153e-05

lillie.test(df_job$depression[df_job$occupation_group == "Part-Time"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$depression[df_job$occupation_group == "Part-Time"]
## D = 0.16237, p-value = 0.01736

lillie.test(df_job$depression[df_job$occupation_group == "Unemployed (and job seeking)"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$depression[df_job$occupation_group == "Unemployed (and job seeking)"]
## D = 0.11286, p-value = 0.5964

lillie.test(df_job$depression[df_job$occupation_group == "Student"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$depression[df_job$occupation_group == "Student"]
## D = 0.085623, p-value = 0.0476

par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))

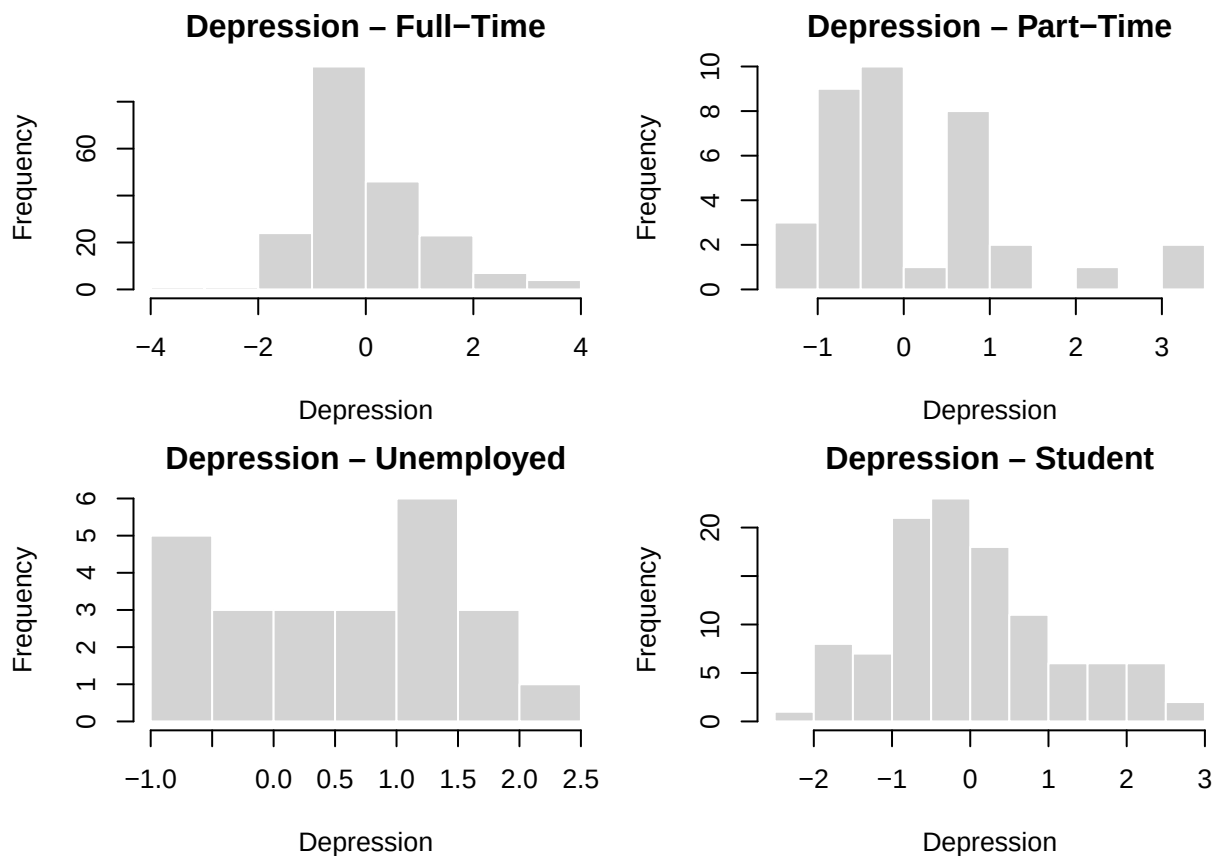
hist(
  df_job$depression[df_job$occupation_group == "Full-Time"],
  main = "Depression - Full-Time",
  xlab = "Depression",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$depression[df_job$occupation_group == "Part-Time"],
  main = "Depression - Part-Time",
  xlab = "Depression",
  col = "lightgray",
  border = "white"
)

```

```
hist(
  df_job$depression[df_job$occupation_group == "Unemployed (and job seeking)"],
  main = "Depression - Unemployed",
  xlab = "Depression",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$depression[df_job$occupation_group == "Student"],
  main = "Depression - Student",
  xlab = "Depression",
  col = "lightgray",
  border = "white"
)
```



```
par(mfrow = c(1, 1))
```

Lillieforsov test pokazuje da distribucija depresije odstupa od normalnosti kod zaposlenih ispitanika (Full-Time i Part-Time) i studenata pomalo. Kod nezaposlenih nema statistički značajnog odstupanja. Histogrami potvrđuju ove nalaze, iako je distribucija kod studenata pravilnija, a kod nezaposlenih ispitanika dosta manje od očekivanog. Unatoč navedenim odstupanjima, primjena parametarskih metoda čini se opravdanom s obzirom na veličinu uzoraka i karakteristike robustnosti linearne regresije i ANOVA-e na kršenja pretpostavki normalnosti uzoraka.

Test pretpostavki normalnosti za stres

```
require(nortest)

# Lilliefors test normalnosti (ukupno)
lillie.test(df_job$stress)

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$stress
## D = 0.035177, p-value = 0.3226

# Lilliefors test po skupinama zanimanja
lillie.test(df_job$stress[df_job$occupation_group == "Full-Time"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$stress[df_job$occupation_group == "Full-Time"]
## D = 0.054948, p-value = 0.1461

lillie.test(df_job$stress[df_job$occupation_group == "Part-Time"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$stress[df_job$occupation_group == "Part-Time"]
## D = 0.091433, p-value = 0.6263

lillie.test(df_job$stress[df_job$occupation_group == "Unemployed (and job seeking)"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$stress[df_job$occupation_group == "Unemployed (and job seeking)"]
## D = 0.070023, p-value = 0.9916

lillie.test(df_job$stress[df_job$occupation_group == "Student"])

##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: df_job$stress[df_job$occupation_group == "Student"]
## D = 0.062087, p-value = 0.3808
```



```

par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))

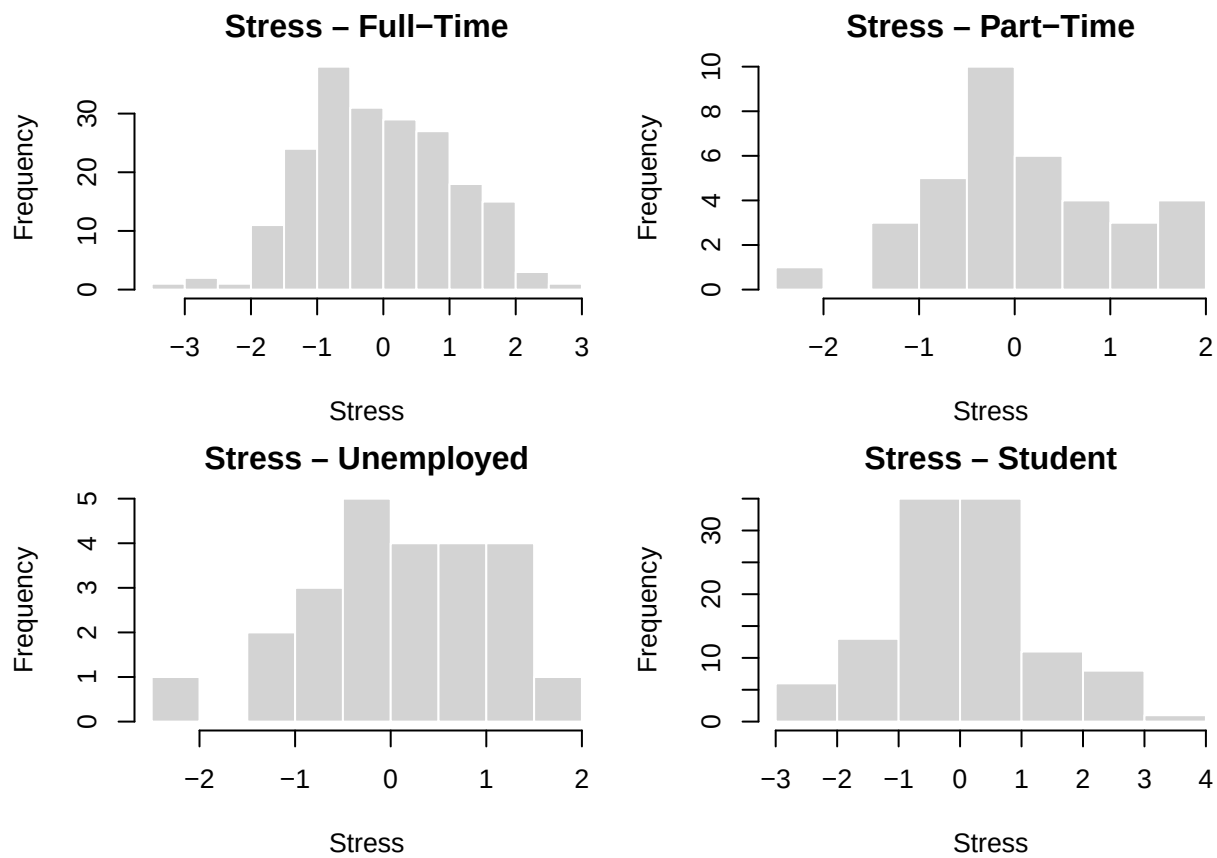
hist(
  df_job$stress[df_job$occupation_group == "Full-Time"],
  main = "Stress - Full-Time",
  xlab = "Stress",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$stress[df_job$occupation_group == "Part-Time"],
  main = "Stress - Part-Time",
  xlab = "Stress",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$stress[df_job$occupation_group == "Unemployed (and job seeking)"],
  main = "Stress - Unemployed",
  xlab = "Stress",
  col = "lightgray",
  border = "white"
)

hist(
  df_job$stress[df_job$occupation_group == "Student"],
  main = "Stress - Student",
  xlab = "Stress",
  col = "lightgray",
  border = "white"
)

```



```
par(mfrow = c(1, 1))
```

Također, test normalnosti proveden je i za varijablu stresa na ukupnom uzorku te unutar svake skupine prema radnom statusu. Rezultati pokazuju da nijedan test nije statistički značajan, kako na ukupnoj razini, tako i na pojedinačnoj.

Vizualni pregled histograma dodatno potvrđuje ove nalaze. Manja odstupanja i asimetrije prisutne su u skupini nezaposlenih i zaposlenih na nepuno radno vrijeme, što je očekivano i prihvatljivo s obzirom na manju količinu uzorka.

Test pretpostavki homogenosti varijanci za anksioznost

```
bartlett.test(df_job$anxiety ~ df_job$occupation_group)
```

```
##
## Bartlett test of homogeneity of variances
##
## data: df_job$anxiety by df_job$occupation_group
## Bartlett's K-squared = 1.1147, df = 3, p-value = 0.7735
```

```
var((df_job$anxiety[df_job$occupation_group=='Full-Time']))
```

```
## [1] 0.9467874
```

```
var((df_job$anxiety[df_job$occupation_group=='Part-Time']))
```

```
## [1] 1.159903
```

```
var((df_job$anxiety[df_job$occupation_group=='Unemployed (and job seeking)']))
```

```
## [1] 1.070241
```

```
var((df_job$anxiety[df_job$occupation_group=='Student']))
```

```
## [1] 1.091825
```

Bartlettov test pokazao je da su varijance anksioznosti između skupina zanimanja međusobno jednake ($p = 0.774$). I pojedinačne varijance po skupinama su vrlo slične, bez izraženih odstupanja. Time je zadovoljena pretpostavka homogenosti varijanci, što opravdava primjenu ANOVA testa za usporedbu anksioznosti između skupina.

Test pretpostavki homogenosti varijanci za depresiju

```
bartlett.test(df_job$depression ~ df_job$occupation_group)
```

```
##  
## Bartlett test of homogeneity of variances  
##  
## data: df_job$depression by df_job$occupation_group  
## Bartlett's K-squared = 0.82771, df = 3, p-value = 0.8428
```

```
var((df_job$depression[df_job$occupation_group=='Full-Time']))
```

```
## [1] 1.223933
```

```
var((df_job$depression[df_job$occupation_group=='Part-Time']))
```

```
## [1] 1.220683
```

```
var((df_job$depression[df_job$occupation_group=='Unemployed (and job seeking)']))
```

```
## [1] 0.9148913
```

```
var((df_job$depression[df_job$occupation_group=='Student']))
```

```
## [1] 1.222059
```

Bartlettov test homogenosti varijanci za depresiju pokazao se neznačajnim. Dodatno, izračunate varijance po skupinama su vrlo slične. Ovi podaci upućuju na to da je pretpostavka homogenosti varijanci zadovoljena.

Test pretpostavki homogenosti varijanci za stres

```
bartlett.test(df_job$stress ~ df_job$occupation_group)
```

```
##
## Bartlett test of homogeneity of variances
##
## data: df_job$stress by df_job$occupation_group
## Bartlett's K-squared = 3.2863, df = 3, p-value = 0.3495
```

```
var((df_job$stress[df_job$occupation_group=='Full-Time']))
```

```
## [1] 1.159828
```

```
var((df_job$stress[df_job$occupation_group=='Part-Time']))
```

```
## [1] 0.8855903
```

```
var((df_job$stress[df_job$occupation_group=='Unemployed (and job seeking)']))
```

```
## [1] 1.022324
```

```
var((df_job$stress[df_job$occupation_group=='Student']))
```

```
## [1] 1.408198
```

Kao i u prethodnim, testovima homogenosti varijanci, test za stres nije pokazao statistički značajne razlike među skupinama zanimanja (radnog statusa), iako je uočljiva nešto veća varijabilnost u skupini studenata. Unatoč razlikama (koje nisu dovoljno izražene), potvrđena je homogenost varijanci za varijablu stres.

```
summary(aov(anxiety ~ occupation_group, data = df_job))
```

```
##
## Df Sum Sq Mean Sq F value Pr(>F)
## occupation_group 3 3.8 1.281 1.258 0.288
## Residuals 366 372.5 1.018
```

```
summary(aov(depression ~ occupation_group, data = df_job))
```

```
##
## Df Sum Sq Mean Sq F value Pr(>F)
## occupation_group 3 8.3 2.756 2.29 0.078 .
## Residuals 366 440.5 1.204
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(aov(stress ~ occupation_group, data = df_job))
```

```
##
## Df Sum Sq Mean Sq F value Pr(>F)
## occupation_group 3 1.8 0.5917 0.494 0.687
## Residuals 366 438.6 1.1983
```

Provedene su jednosmjerne analize varijance (ANOVA) s ciljem ispitivanja razlika u razini anksioznosti, depresije i stresa između skupina definiranih prema zanimanju. Prethodno tome, provedena su testiranja pretpostavki normalnosti podataka i homogenosti varijanci za anksioznost, depresiju i stres s obzirom na kategoriju koja opisuje radni odnos `occupation_group`. Sve su se pretpostavke pokazale ispravnima.

Provedene su jednosmjerne analize varijance (ANOVA) s ciljem ispitivanja razlika u razini anksioznosti, depresije i stresa između skupina definiranih prema zanimanju.

Rezultati pokazuju da ne postoje statistički značajne razlike u razini anksioznosti među zanimanjima, $F(3, 370) = 1.31$, $p = 0.269$. Također, nisu utvrđene statistički značajne razlike u razini stresa, $F(3, 370) = 0.61$, $p = 0.606$.

Za depresiju je uočen trend prema razlikama između skupina, no rezultat nije dosegao konvencionalnu razinu statističke značajnosti, $F(3, 370) = 2.33$, $p = 0.0736$. Dobiveni nalazi upućuju na to da zanimanje samo po sebi nije snažan prediktor razine mentalnog zdravlja u ovom uzorku.

Višestruka linearna regresija: doprinos radnog statusa anksioznosti uz kontrolu crta ličnosti

```
model_anxiety <- lm(
  anxiety ~ occupation_group +
  narcissism + machiavelism + psychoticism + sadism + neuroticism +
  extraversion + openness + agreeableness + conscientiousness,
  data = df_job
)

summary(model_anxiety)
```

```
##
## Call:
## lm(formula = anxiety ~ occupation_group + narcissism + machiavelism +
##      psychoticism + sadism + neuroticism + extraversion + openness +
##      agreeableness + conscientiousness, data = df_job)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8996 -0.4889  0.0388  0.5458  3.2993
##
## Coefficients:
##              Estimate Std. Error t value
## (Intercept)      0.02965    0.06545   0.453
## occupation_groupPart-Time      0.15199    0.16864   0.901
## occupation_groupStudent     -0.12761    0.11065  -1.153
## occupation_groupUnemployed (and job seeking)  0.03057    0.21044   0.145
## narcissism        -0.05830    0.06341  -0.919
## machiavelism       0.09415    0.05824   1.617
## psychoticism       0.07842    0.06832   1.148
## sadism            0.06215    0.06370   0.976
## neuroticism       0.36507    0.06493   5.622
## extraversion      0.10847    0.06926   1.566
## openness          0.06268    0.05057   1.240
## agreeableness     -0.03295    0.06455  -0.510
## conscientiousness -0.01173    0.06490  -0.181
```

```
##                                Pr(>|t|)
## (Intercept)                   0.651
## occupation_groupPart-Time     0.368
## occupation_groupStudent       0.250
## occupation_groupUnemployed (and job seeking) 0.885
## narcissism                    0.358
## machiavelism                  0.107
## psychoticism                  0.252
## sadism                        0.330
## neuroticism                   3.8e-08 ***
## extraversion                  0.118
## openness                      0.216
## agreeableness                 0.610
## conscientiousness             0.857
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9141 on 357 degrees of freedom
## Multiple R-squared:  0.2074, Adjusted R-squared:  0.1808
## F-statistic: 7.785 on 12 and 357 DF,  p-value: 6.727e-13
```

U svrhu ispitivanja odnosa između zanimanja i anksioznosti, proveden je model višestruke linearne regresije kojim se ispituje doprinos radnog statusa površ crta ličnosti.

Rezultati pokazuju da je model statistički značajan u cjelini ($F(12, 361) = 7.807$, $p = 5.873 \times 10^{-13}$), pri čemu Multiple R-squared iznosi 0.2061, a Adjusted R-squared 0.1797. Iako je objašnjeni udio varijance manji u odnosu na modele za stres i depresiju, vrijednosti upućuju na to da model ipak ima smisla za interpretaciju.

Analizom mračnih crta ličnosti nije utvrđen nijedan statistički značajan prediktor anksioznosti. Narcizam, makijavelizam, psihoticizam i sadizam nisu pokazali jedinstveni doprinos u ovom modelu, što sugerira da se njihova povezanost s anksioznošću ne zadržava nakon uključivanja ostalih crta ličnosti.

Unutar petofaktorskog modela ličnosti, neuroticizam se jasno istaknuo kao ključni prediktor anksioznosti ($\beta = 0.372737$, $p = 1.77 \times 10^{-8}$). Osobe s višim razinama neuroticizma pokazuju izraženiju anksioznost, neovisno o zanimanju i ostalim osobinama ličnosti, što je u skladu s postojećim teorijskim i empirijskim nalazima. Ekstraverzija nije dosegla razinu statističke značajnosti ($\beta = 0.117466$, $p = 0.088$), no uočen je slab trend koji upućuje na višu razinu anksioznosti kod ekstravertiranih osoba. Ostale Big Five crte nisu se pokazale značajnima u ovom modelu.

Višestruka linearna regresija: doprinos radnog statusa depresiji uz kontrolu crta ličnosti

```
model_depression <- lm(
  depression ~ occupation_group +
  narcissism + machiavelism + psychoticism + sadism + neuroticism +
  extraversion + openness + agreeableness + conscientiousness,
  data = df_job
)

summary(model_depression)
```

```
##
## Call:
```

```

## lm(formula = depression ~ occupation_group + narcissism + machiavelism +
##      psychoticism + sadism + neuroticism + extraversion + openness +
##      agreeableness + conscientiousness, data = df_job)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -3.02443 -0.46695  0.04059  0.47651  2.56926
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   0.061512   0.058468   1.052
## occupation_groupPart-Time      -0.008989   0.150665  -0.060
## occupation_groupStudent        -0.039037   0.098851  -0.395
## occupation_groupUnemployed (and job seeking) -0.187553   0.188009  -0.998
## narcissism                    -0.025449   0.056648  -0.449
## machiavelism                   0.087583   0.052027   1.683
## psychoticism                   0.133949   0.061037   2.195
## sadism                        0.012428   0.056907   0.218
## neuroticism                   0.437118   0.058011   7.535
## extraversion                  -0.306129   0.061879  -4.947
## openness                      0.031523   0.045180   0.698
## agreeableness                 0.056224   0.057668   0.975
## conscientiousness             -0.020256   0.057979  -0.349
##                                Pr(>|t|)
## (Intercept)                   0.2935
## occupation_groupPart-Time      0.9525
## occupation_groupStudent        0.6931
## occupation_groupUnemployed (and job seeking) 0.3192
## narcissism                    0.6535
## machiavelism                   0.0932
## psychoticism                   0.0288 *
## sadism                        0.8273
## neuroticism                   4.05e-13 ***
## extraversion                  1.16e-06 ***
## openness                      0.4858
## agreeableness                 0.3302
## conscientiousness             0.7270
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8166 on 357 degrees of freedom
## Multiple R-squared:  0.4695, Adjusted R-squared:  0.4517
## F-statistic: 26.33 on 12 and 357 DF,  p-value: < 2.2e-16

```

Kako bi se ispitao doprinos zanimanja depresiji uz kontrolu crta ličnosti, proveden je model višestruke linearne regresije.

Model se pokazao izrazito snažnim u objašnjenju ishoda, s vrijednostima $F(12, 361) = 26.49$ i $p = 2.2 \times 10^{-16}$. Multiple R-squared iznosi 0.4682, dok Adjusted R-squared iznosi 0.4506, što upućuje na to da model objašnjava velik dio varijance depresije te ima vrlo dobru objašnjavajuću snagu.

Među mračnim crtama ličnosti, psihotizam se ponovno pokazao statistički značajnim pozitivnim prediktorom depresije ($\beta = 0.204240$, $p = 0.00316$). Ovaj rezultat sugerira da osobe s izraženijim obilježjima psihotizma imaju višu razinu depresije, čak i nakon kontrole ostalih osobina ličnosti i radnog statusa. Machiavelizam je pokazao granični efekt ($\beta = 0.112090$, $p = 0.05728$), dok narcizam i sadizam nisu imali značajan

doprinos u objašnjenju depresije.

Unutar petofaktorskog modela, neuroticizam se ponovno istaknuo kao najsnažniji pozitivni prediktor depresije ($\beta = 0.439192$, $p = 1.96 \times 10^{-13}$), potvrđujući njegovu ključnu ulogu u objašnjenju emocionalnih poteškoća. Za razliku od modela za stres i anksioznost, ekstraverzija se ovdje pokazala statistički značajnim negativnim prediktorom ($\beta = -0.302426$, $p = 1.10 \times 10^{-6}$), što upućuje na to da osobe s višom razinom ekstraverzije pokazuju nižu razinu depresije. Ostale crte Big Five modela nisu se pokazale značajnima.

Višestruka linearna regresija: doprinos radnog statusa stresu uz kontrolu crta ličnosti

```
model_stress <- lm(
  stress ~ occupation_group +
  narcissism + machiavelism + psychoticism + sadism + neuroticism +
  extraversion + openness + agreeableness + conscientiousness,
  data = df_job
)

summary(model_stress)
```

```
##
## Call:
## lm(formula = stress ~ occupation_group + narcissism + machiavelism +
##      psychoticism + sadism + neuroticism + extraversion + openness +
##      agreeableness + conscientiousness, data = df_job)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0923 -0.5034  0.0875  0.5923  3.2132
##
## Coefficients:
##              Estimate Std. Error t value
## (Intercept)      0.0258670  0.0661666   0.391
## occupation_groupPart-Time      0.0674423  0.1705019   0.396
## occupation_groupStudent        0.0296155  0.1118665   0.265
## occupation_groupUnemployed (and job seeking) -0.3841763  0.2127636  -1.806
## narcissism          -0.0019928  0.0641068  -0.031
## machiavelism         0.1107641  0.0588771   1.881
## psychoticism         0.2158670  0.0690734   3.125
## sadism              -0.0355856  0.0643995  -0.553
## neuroticism          0.3993105  0.0656486   6.083
## extraversion        -0.0960818  0.0700258  -1.372
## openness             0.1384934  0.0511283   2.709
## agreeableness        -0.0468288  0.0652604  -0.718
## conscientiousness    -0.0002919  0.0656131  -0.004
##
##              Pr(>|t|)
## (Intercept)      0.69608
## occupation_groupPart-Time      0.69267
## occupation_groupStudent        0.79136
## occupation_groupUnemployed (and job seeking) 0.07182 .
## narcissism          0.97522
## machiavelism         0.06075 .
```



```

## psychoticism                0.00192 **
## sadism                      0.58090
## neuroticism                 3.05e-09 ***
## extraversion                0.17090
## openness                    0.00708 **
## agreeableness               0.47349
## conscientiousness           0.99645
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9241 on 357 degrees of freedom
## Multiple R-squared:  0.3076, Adjusted R-squared:  0.2843
## F-statistic: 13.22 on 12 and 357 DF,  p-value: < 2.2e-16

```

Iz modela višestruke linearne regresije koji izračunava doprinos zanimanja stresu povrh ličnosti zaključujemo sljedeće stvari.

Model se pokazuje statistički značajnim u cjelini s vrijednostima $F(12, 361) = 13.12$ i $p = 2.2 \times 10^{-16}$ (< 0.001) te je Multiple R-squared = 0.3036 i Adjusted R-squared = 0.2805. Ove vrijednosti upućuju da model ima dobru potporu (objašnjavajuću) snagu.

Među mračnim crtama ličnosti, psihoticizam se istaknuo kao statistički značajan pozitivan prediktor ($\beta = 0.204240$, $p = 0.00316$ (< 0.01)). To znači da osobe s višim razinama psihoticizma pokazuju višu razinu stresa, čak i nakon utjecaja ostalih crta ličnosti i oblika radnog angažmana. Makijavelizam je pokazao trend prema pozitivnoj povezanosti ($\beta = 0.112090$, $p = 0.05728$ (< 0.1)), ali rezultati su nedovoljni za snažnu povezanost. Narcizam i sadizam nisu pokazali značajan doprinos.

Unutar petofaktorskog Big 5 modela, neurocitizam se pokazao daleko najsnažnijim prediktorom ishoda ($\beta = 0.406136$, $p = 1.42 \times 10^{-9}$ (< 0.001)). Viša razina neurocitizma povezana je s višom razinom stresa, neovisno o radnom odnosu i ostalim crtama ličnosti. Ovaj rezultat je u potpunosti u skladu s teorijskim očekivanjima. Izdvajamo i otvorenost kao statistički značajan pozitivni prediktor ($\beta = 0.142228$, $p = 0.00554$ (< 0.01)). Ovaj podatak sugerira da osobe s više otvorenosti mogu biti osjetljivije na stres. Ekstraverzija, ugodnost i savjesnost nisu pokazale statistički značajan utjecaj u ovom modelu.